RS5 — The microwave background and the growth of structure on the deflating spacetimes: perturbation equations, closures, spectra and forecasts

PROJECT spacetimes, Revision_Deflation. Supplement to both manuscripts of the revision

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0. Purpose and summary

The supplement S3 of the main project computes the temperature and polarization spectra of the cosmic microwave background and the power spectrum of matter with an Einstein–Boltzmann program of its own, and uses them three times: to validate the program, to forecast how well mock data constrain its families of dark radiation, and to follow the growth of structure when the dark matter is a superfluid. The task of the revision is to repeat every calculation of the main project while the coordinates $x5$, $x6$, $x7$ deflate. This supplement repeats those three.

The supplement RS3 treats the *unperturbed* universe on the deflating spacetimes: if gravity is eight-dimensional (assumption A2 of RS3), Newton's constant grows at the total rate of deflation, gravity was weaker in the past, and the Hubble constant inferred from the sound horizon falls. The spectra need more than that. They are built from small perturbations of ordinary space, and the equations that govern the perturbations contain the *perturbed* stresses of whatever keeps the three rates of deflation constant. The author's hypotheses prescribe the metric of the deflating coordinates; they do not say what holds it. One further assumption is therefore unavoidable. It is called A3 here, it is stated in Sec. 1, and its consequences are kept apart from those of A1 and A2 throughout.

Summary.

1. *The equations* (Sec. 2). With the scale factors of the extra coordinates unperturbed, the Einstein equations of ordinary space to first order in a perturbation are those of four dimensions with Newton's constant of the moment and with the conformal Hubble rate H_{cf} replaced by $H_e = H_{\mathrm{cf}} - a_{\mathrm{sp}}\,\mathrm{gsum}/2$ in three of them. This is derived by computer algebra for arbitrary rates and both signatures.
2. *The four equations cannot all hold with matter alone* (Sec. 3). In general relativity two of the four scalar Einstein equations imply the other two. With a Newton's constant that depends on the time this fails by terms proportional to gsum: the agent that holds the extra coordinates must carry perturbations, and one must say which. Each choice is a *closure*. Two are computed ("Momentum", "Pressure"); a third ("Shear") is shown to coincide with "Pressure" far inside the horizon.
3. *What does not depend on the closure* (Sec. 7). For the multipoles $\ell \geq 30$, which carry almost all the information, the spectra of the two closures agree to a thousandth of the effect of the deflation: they follow from A1, A2 and the first two parts of A3, whatever the closure. At the rate that lunar laser ranging allows, mock data of the quality of the Planck satellite cannot tell the spectra from those without deflation; at the rate that nucleosynthesis allows they differ by up to 5.6 per cent.
4. *What does depend on the closure* (Secs. 4, 7.3). The growth of structure and the largest angular scales ($\ell < 30$) depend on the closure at the first order in gsum/H_0. At a fixed Hubble constant

the amplitude of clustering falls by $0.443\,\mathrm{gsum}/H_0$ in one closure and by $0.207\,\mathrm{gsum}/H_0$ in the other. These are not predictions of the hypotheses. An earlier version of the revision quoted a single number, from an equation that is neither closure; that was an error (provenance RP9, item R-E33).

5. *The Hubble constant* (Secs. 6, 8). With recombination computed on the deflating background, the Hubble constant inferred at a fixed angle of the sound horizon falls about six per cent faster than RS3 first reported: $d\ln H_0/d(\mathrm{gsum}/H_0) = -0.969$. In fits of all six parameters of the standard model to mock data it falls faster still, by 84 km/s/Mpc per unit of gsum/H_0. Deflation cannot raise the Hubble constant inferred from the microwave background.
6. *A new bound, as a forecast* (Sec. 8). Planck-like mock data bound the rate to $\mathrm{gsum} < 0.0385\,H_0$ at 95 per cent, and to $0.0207\,H_0$ with a deep survey added: weaker than lunar laser ranging by a factor of 12, stronger than nucleosynthesis by a factor of 1.7.
7. *The dark-radiation families of the main project* (Sec. 9) are fitted again on the deflating background. At the rate that lunar laser ranging allows, every best-fit Hubble constant of the main project falls by about a quarter of a km/s/Mpc, and its conclusions stand. With the rate free, more extra radiation is allowed, because extra radiation and weaker gravity act against each other in the early universe; but the Hubble constant falls along that compensation as well, and the largest Hubble constant that the mock data allow is reached at zero rate for every family.
8. *The superfluid dark matter of the main project* (Sec. 10) is followed again on the deflating background. The scale below which its pressure suppresses structure moves by 0.15 to 0.16 per cent at the rate that lunar laser ranging allows and by about 3 per cent at the rate that nucleosynthesis allows, in either closure; the limits on the fermion mass and on the sound speed of a condensate move accordingly.

0.1 How to reproduce

```
cd "/Users/nsh/Developer/github/spacetimes/Revision_Deflation"
python3 code/scripts/vendor_eb.py          # writes the four copied modules (eb_*.rs)
cargo build --release --manifest-path code/deflation-rs/Cargo.toml
cargo test --release --manifest-path code/deflation-rs/Cargo.toml
./code/deflation-rs/target/release/deflation cosmology    # RS3 Sec. 2; Sec. 6 here
./code/deflation-rs/target/release/deflation matter       # Sec. 4
./code/deflation-rs/target/release/deflation background   # Sec. 5.2      (seconds)
./code/deflation-rs/target/release/deflation spectrum     # Sec. 7        (1 minute)
./code/deflation-rs/target/release/deflation forecast     # Sec. 8        (6 minutes)
./code/deflation-rs/target/release/deflation families     # Sec. 9        (40 minutes)
./code/deflation-rs/target/release/deflation superfluid   # Sec. 10       (20 minutes)
python3 code/scripts/verify_deflating_independent.py  # independent checks (I44d-, I71d-, Idef-)
python3 code/scripts/mutation_test_independent.py         # can these checks fail? (4 minutes)
python3 code/scripts/make_figures.py spectra closure forecast families sfdm hubble bounds
python3 code/scripts/write_doc_tables.py
python3 code/scripts/assemble_docs.py docs_src/RS5_microwave_background_deflating.src.md
bash code/scripts/build_docs.sh supplement/RS5_microwave_background_deflating.md
```

(The times are for two processor cores.)

0.2 Symbols

The symbols of this supplement that are new, each defined in words where it first appears below:

symbol	in programs and data files	meaning	defined in	origin
τ_{cf}	`tau_cf`, `tau0_Mpc`	the conformal time of ordinary space, $d\tau_{\mathrm{cf}} = dx4/a_{\mathrm{sp}}$, in Mpc (`tau0_Mpc`: its value today). In the equations of the perturbations a prime is the derivative with respect to τ_{cf}. (The letter τ without index is the optical depth of the main registry, and τ_d is the deflation clock)	RS5 2; M XI	new

symbol	in programs and data files	meaning	defined in	origin
$H_{\rm cf}$, H_e	`Hc`, `He`, `hub`, `hub_e`, `a_gsum_over_Hc`	the conformal Hubble rate $H_{\rm cf} = a_{\rm sp} H$, and $H_e = H_{\rm cf} - a_{\rm sp}\,\mathrm{gsum}/2$, which takes its place in three of the Einstein equations of ordinary space when the extra coordinates deflate	RS5 2; M XI	new
k (perturbations), ℓ, D_ℓ^{TT}, D_ℓ^{TE}, D_ℓ^{EE}	—	the comoving wavenumber of a perturbation of ordinary space (in Mpc^{-1}; the physical wavenumber is $k/a_{\rm sp}$); the multipole; and the spectra of the temperature, of the polarization and their cross spectrum, $D_\ell = \ell(\ell+1)C_\ell/2\pi$ in $\mu\mathrm{K}^2$, as in the supplement S3 of the main project	RS5 2, 7; M XI	new
$h_{\rm sy}$, $\eta_{\rm sy}$, $\alpha_{\rm sy}$	—	the two functions that describe a scalar perturbation of the metric of ordinary space in the synchronous gauge (the h and η of Ma and Bertschinger; here with the index sy, because h is the Hubble parameter and η the flat metric), and the combination $\alpha_{\rm sy} = (h'_{\rm sy} + 6\eta'_{\rm sy})/2k^2$	RS5 2; M XI	new
$\mathcal{U}$	`potential`	$\eta_{\rm sy} + \alpha'_{\rm sy}$, the sum of the two gravitational potentials of the conformal Newtonian gauge (script letter): the potential that light feels	RS5 2, 7	new
$\mathcal{D}$, $\mathcal{T}$, $\mathcal{P}$, $\mathcal{A}$	`d_sum`, `three_dpres`	$8\pi G_N a_{\rm sp}^2$ times the perturbation of the density, of the divergence of the momentum density, of the pressure, and of the anisotropic stress of matter and radiation together (script letters; in the notation of Ma and Bertschinger: $\delta\rho$, $(\rho+P)\theta$, δP, $(\rho+P)\sigma$)	RS5 2	new
$\mathcal{E}$, $\mathcal{E}_P$, $\mathcal{E}_m$, $\mathcal{E}_r$, $\mathcal{B}$	`grho_sum`, `half_rho_plus_p`, `b_trace`, `n_bg`	$8\pi G_N a_{\rm sp}^2$ times the density and the pressure of the unperturbed universe, summed over the species ($\mathcal{E}_m$, $\mathcal{E}_r$: the parts of $\mathcal{E}$ from pressureless matter and from radiation); and $\mathcal{B} = H'_e - H_e^2 + \frac{1}{2}(\mathcal{E} + \mathcal{E}_P)$, which vanishes without deflation (script letters)	RS5 2, 3, 4	new
$p_{\rm agent}$	—	the pressure in ordinary space of the agent, that is, of whatever keeps the three rates of deflation constant, in the unperturbed universe (under assumption A2 the agent has no energy density)	RS5 1	new
$\mathcal{Y}_M$, $\mathcal{Y}_T$, $\mathcal{Y}_S$; $\mathcal{Z}$	`momentum_gap`, `trace_gap`, `shear_gap`, `x_gap`, `shear_check`, `trace_check`	the gaps: what the momentum constraint, the trace equation and the equation of the shear lack when only matter and radiation stand on their right-hand sides; the agent that holds the extra scale factors supplies them. $\mathcal{Z} = h'_{\rm sy}\,\mathcal{B} - \frac{1}{2} a_{\rm sp}\,\mathrm{gsum}\,(\mathcal{D} - 3\mathcal{P})$ (script letters)	RS5 3	new
$Y_{\rm He}$	—	the mass fraction of helium after nucleosynthesis (with the index He; `Y_He` in the data files, as in the main project)	RS5 5; M XI	new
τ (main registry)	`tau_reio`	the optical depth of reionization of the main registry; `tau_reio` is its name in the data files that the revision writes (the bare name `tau` is not used there)	RS5 8	main

The symbols of RS3 (G_N, $a_{\rm sp}$, H, θ_*, z_*, δ_m, f_g, σ_8) and of the main registry (ω_b, ω_c, h, H_0, A_s, n_s, τ, N, z_t, $\Delta\chi^2$, r_s, c_{s0}, R_{TF}) keep their meanings. *A warning about letters.* The standard reference for the equations of this supplement, Ma and Bertschinger (Ref. [1]), writes h, η, α, θ, σ and τ for quantities that have other meanings in this project (h is the Hubble parameter, η the flat metric, τ the optical depth, and so on). This supplement therefore writes $h_{\rm sy}$, $\eta_{\rm sy}$, $\alpha_{\rm sy}$ and $\tau_{\rm cf}$, and uses script letters for the sums over matter and radiation.

1. What must be assumed

1.1 The unperturbed universe, and the agent

RS3 makes two assumptions. **A1**: ordinary space $x1, x2, x3$ expands with the scale factor $a_{\rm sp}(x4)$ and the Hubble rate $H = d\ln a_{\rm sp}/dx4$, while $x5, x6, x7$ deflate at the constant rates g55, g66, g77 and the hidden coordinate $x0$ is static. **A2**: the metric obeys Einstein's equations in eight dimensions with a constant coupling G_8, and the component of those equations along the time contains matter and radiation only. Two consequences follow (RS3, Sec. 1). Newton's constant of ordinary space is G_8 divided by the proper volume of the extra coordinates and grows as

$$G_N(x4) = G_N(\text{today})\, e^{\,\text{gsum}\,(x4 - x4_{\text{today}})}, \qquad \text{gsum} = \text{g55} + \text{g66} + \text{g77}, \tag{1.1}$$

and the expansion obeys

$$3H^2 - 3\,\text{gsum}\,H + \text{gprod} = 8\pi\, G_N(x4)\,\rho, \qquad \text{gprod} = \text{g55}\,\text{g66} + \text{g55}\,\text{g77} + \text{g66}\,\text{g77}, \tag{1.2}$$

with the density ρ of matter, radiation and the cosmological term.

Something must keep the three rates constant while ordinary space expands and matter dilutes; left to themselves the extra scale factors would obey equations of their own. This supplement calls that something *the agent*. Nothing is assumed about its nature. What A2 says about it is that it has no energy density: otherwise its density would stand on the right of Eq. (1.2). It cannot be without stresses, however. The components of Einstein's equations along ordinary space read, for the unperturbed universe,

$$-\Big(2\,\frac{dH}{dx4} + 3H^2 - 2\,\text{gsum}\,H + \text{gsq} + \text{gprod}\Big) = 8\pi G_N\,(P + p_{\text{agent}}), \qquad \text{gsq} = \text{g55}^2 + \text{g66}^2 + \text{g77}^2,$$

with the pressure P of matter and radiation and the pressure p_{agent} of the agent in ordinary space. Matter is conserved in ordinary space, so that $d(8\pi G_N\rho)/dx4 = \text{gsum}\,8\pi G_N\rho - 3H\,8\pi G_N(\rho + P)$ by Eq. (1.1); the derivative of Eq. (1.2) then gives P, and what remains is

$$8\pi G_N\, p_{\text{agent}} = -\frac{\text{gsum}}{H}\frac{dH}{dx4} - 2\,\text{gsum}\,H + 2\,\text{gprod} - \frac{\text{gsum}\,\text{gprod}}{3H} \tag{1.3}$$

(check `Idef-P8`). In an era of matter, where $dH/dx4 = -\frac{3}{2}H^2$, this is $-\text{gsum}\,H/2$ to the first order in the rates: **the agent is a medium without energy density and with a tension** of the relative size gsum/H. (In an era of radiation, $dH/dx4 = -2H^2$, the first two terms cancel and the pressure vanishes to this order; when the cosmological term dominates it is $-2\,\text{gsum}\,H$.) (It also has stresses along $x0$ and along the deflating coordinates, which do not enter what follows.) When ordinary space is perturbed, such a medium will in general be perturbed too.

1.2 The third assumption

A3 — the perturbations.

- *(i) The extra scale factors are not perturbed.* In a universe with small inhomogeneities the scale factors b_5, b_6, b_7 of the deflating coordinates remain the prescribed functions of the time, and $x0$ remains static. "The time" is the proper time of observers who move with the cold matter (the time coordinate of the synchronous gauge, Sec. 2.1). This is the plainest reading of the author's formula $g_{55} = \pm\,\text{const5}^2\, e^{-2\,\text{g55}\,x4}$ for the component g_{55} of the metric in a universe that is not exactly homogeneous.
- *(ii) The agent has no energy density in the perturbed universe either.* This is assumption A2 taken to first order.
- *(iii) A closure.* One more statement about the perturbations of the agent is needed (Sec. 3). The hypotheses offer no ground for a choice. The program computes with two closures, and every result is marked as independent of the closure or as depending on it.

Parts (i) and (ii) are assumptions of this supplement, not hypotheses of the author. A different part (i) — for instance scale factors that respond to the local density — would give a different theory of the perturbations, which is not examined here.

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2. The Einstein equations of ordinary space to first order

2.1 The perturbed metric

The calculation uses the conformal time $\tau_{\rm cf}$ of ordinary space, defined by $d\tau_{\rm cf} = dx4/a_{\rm sp}$, and writes a prime for the derivative with respect to it. The conformal Hubble rate is $H_{\rm cf} = a'_{\rm sp}/a_{\rm sp} = a_{\rm sp}H$. A scalar perturbation of ordinary space with the comoving wavenumber k along $x3$ is described, in the synchronous gauge of Ref. [1], by two functions $h_{\rm sy}(\tau_{\rm cf})$ and $\eta_{\rm sy}(\tau_{\rm cf})$:

$$ds^2 = \eta_{00}\,dx0^2 + \eta_{11}\,a_{\rm sp}^2\Big[(1-2\eta_{\rm sy}\cos kx3)(dx1^2+dx2^2) + (1+(h_{\rm sy}+4\eta_{\rm sy})\cos kx3)dx3^2\Big] + \eta_{44}\,a_{\rm sp}^2\,d\tau_{\rm cf}^2 + \eta_{55}\sum_{t=5}^{7} b_t^2\,dxt^2, \tag{2.1}$$

where η_{00}, η_{11}, η_{44}, η_{55} are the signs of the flat metric of the spacetime at hand: $+1,+1,-1,-1$ on the 4+4 spacetime and $-1,-1,+1,-1$ on the 7+1 spacetime. The results below are the same for both (checks `I44d-P1`, `I71d-P1`). The components g_{55}, g_{66}, g_{77} of the metric are $\eta_{55}\,b_t^2$ with the scale factors $b_t = \text{const}_t\,e^{-\text{rate}_t\,x4}$, which by part (i) of A3 are not perturbed. "Synchronous" means that the time-time and time-space components of the metric are not perturbed: the coordinate time is the proper time of freely falling observers who were at rest initially, which are the particles of cold matter. The combination

$$\alpha_{\rm sy} = \frac{h'_{\rm sy} + 6\,\eta'_{\rm sy}}{2k^2} \tag{2.2}$$

appears repeatedly; the sum of the two gravitational potentials that light feels (the two potentials of the conformal Newtonian gauge) is $\mathcal{U} = \eta_{\rm sy} + \alpha'_{\rm sy}$.

2.2 The four equations

Matter and radiation enter through four sums over the species, each multiplied by $8\pi G_N(x4)\,a_{\rm sp}^2$ with Newton's constant of the moment, Eq. (1.1):

- $\mathcal{D}$: the perturbation of the energy density;
- $\mathcal{T}$: the divergence of the momentum density (in Ref. [1]: $(\rho+P)\theta$);
- $\mathcal{P}$: the perturbation of the pressure;
- $\mathcal{A}$: the anisotropic stress (in Ref. [1]: $(\rho+P)\sigma$).

The Einstein tensor of the metric (2.1) to first order, computed by computer algebra for arbitrary rates (checks `I44d-P1`, `I71d-P1`), gives

$$\begin{aligned}
&\text{(E)} && 2k^2\eta_{\rm sy} - H_e\,h'_{\rm sy} = -\mathcal{D}, && \text{energy constraint (time-time)}\\
&\text{(M)} && 2k^2\eta'_{\rm sy} = \mathcal{T}, && \text{momentum constraint (time-space)}\\
&\text{(T)} && h''_{\rm sy} + 2H_e\,h'_{\rm sy} - 2k^2\eta_{\rm sy} = -3\,\mathcal{P}, && \text{trace of the space-space part}\\
&\text{(S)} && \alpha'_{\rm sy} + 2H_e\,\alpha_{\rm sy} - \eta_{\rm sy} = -\tfrac{3}{2}\,\mathcal{A}/k^2, && \text{traceless space-space part (shear)}
\end{aligned} \tag{2.3}$$

with

$$H_e = H_{\rm cf} - \tfrac{1}{2}\,a_{\rm sp}\,\text{gsum}. \tag{2.4}$$

These are the equations (21) of Ref. [1] with two changes: Newton's constant is that of the moment, and $H_{\rm cf}$ is replaced by H_e in (E), (T) and (S). The reason for the second change can be read off Eq. (1.2). The time-time component of the Einstein tensor is the sum of the products of all pairs of expansion rates. If the rate of one ordinary direction is perturbed, the perturbation multiplies the sum of the rates of all *other* directions: $2H$ in four dimensions, $2H-$gsum in eight dimensions with three deflating coordinates. In the same way the friction term of a shear, which in four dimensions is the sum $3H$ of all rates, becomes $3H-$ gsum. The momentum constraint contains no expansion rate and is unchanged.

The Einstein tensor has perturbed components along $x0$ and along the deflating coordinates as well (same checks). They are the stresses with which the agent holds those coordinates, and they are not needed below.

2.3 Matter and radiation

Matter and radiation that are spread uniformly over the extra coordinates obey the four-dimensional equations of motion unchanged: their energy and momentum are conserved in ordinary space, a photon without momentum along the extra coordinates follows a geodesic of the metric of ordinary spacetime, and Thomson scattering and recombination are atomic physics, which the model does not alter (RS3, Sec. 3, item 6). The program therefore keeps every equation of motion of the main project for photons, neutrinos, baryons, cold dark matter and the dark fluids (S3, Sec. 2.1; Ref. [1]) and changes only what multiplies Newton's constant and the expansion rate. Two relations of the unperturbed universe are used throughout. With $\mathcal{E}=8\pi G_N a_{\rm sp}^2\rho$ and $\mathcal{E}_P=8\pi G_N a_{\rm sp}^2 P$, summed over the species, Eq. (1.2) reads

$$H_{\rm cf}^2-a_{\rm sp}\,\text{gsum}\,H_{\rm cf}+\tfrac{1}{3}a_{\rm sp}^2\,\text{gprod}=\tfrac{1}{3}\,\mathcal{E}, \tag{2.5}$$

and each species with the equation of state $P=w\rho$ obeys

$$\left(8\pi G_N a_{\rm sp}^2\rho\right)'=\left(a_{\rm sp}\,\text{gsum}-(1+3w)H_{\rm cf}\right)8\pi G_N a_{\rm sp}^2\rho, \tag{2.6}$$

in which the term $a_{\rm sp}\,$gsum is the growth of Newton's constant. The derivative of $H_{\rm cf}$ follows from the two (check `Idef-P2`).

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3. Why the four equations cannot all hold: gaps and closures

3.1 Two identities

In general relativity the four equations (2.3) are not independent. Because energy and momentum of matter are conserved, the derivative of (E) is a combination of (M) and (T), and the derivative of (M) a combination of (T) and (S): two equations determine the two functions $h_{\rm sy}$, $\eta_{\rm sy}$, and the other two hold automatically. The same manipulation on the deflating spacetime shows what is different. Define the three *gaps*, the amounts by which (M), (T) and (S) fail when only matter and radiation stand on their right-hand sides:

$$\mathcal{Y}_M=2k^2\eta_{\rm sy}'-\mathcal{T},\qquad \mathcal{Y}_T=h_{\rm sy}''+2H_e\,h_{\rm sy}'-2k^2\eta_{\rm sy}+3\,\mathcal{P},\qquad \mathcal{Y}_S=\alpha_{\rm sy}'+2H_e\,\alpha_{\rm sy}-\eta_{\rm sy}+\tfrac{3}{2}\,\mathcal{A}/k^2. \tag{3.1}$$

First identity. Differentiate (E). The derivative of $\mathcal{D}$ follows from Eq. (2.6) and from the conservation of energy of each species (Ref. [1], Eq. (30)): $\mathcal{D}'=(a_{\rm sp}\,\text{gsum}-H_{\rm cf})\,\mathcal{D}-3H_{\rm cf}\,\mathcal{P}-\mathcal{T}-\frac{1}{2}(\mathcal{E}+\mathcal{E}_P)\,h_{\rm sy}'$. Eliminating $h_{\rm sy}''$ with the definition of $\mathcal{Y}_T$ and $\eta_{\rm sy}$ with (E) leaves

$$\mathcal{Y}_M=H_e\,\mathcal{Y}_T+\mathcal{Z},\qquad \mathcal{Z}=h_{\rm sy}'\,\mathcal{B}-\tfrac{1}{2}\,a_{\rm sp}\,\text{gsum}\,(\mathcal{D}-3\,\mathcal{P}), \tag{3.2}$$

with a quantity of the unperturbed universe,

$$\mathcal{B} = H_e' - H_e^2 + \tfrac{1}{2}(\mathcal{E} + \mathcal{E}_P) = \frac{a_{\rm sp}\,\mathrm{gsum}}{2H_e}\Big[\tfrac{1}{6}\textstyle\sum(1-3w)\,8\pi G_N a_{\rm sp}^2\rho + \tfrac{1}{4}a_{\rm sp}^2\,\mathrm{gsum}^2 - \tfrac{1}{3}a_{\rm sp}^2\,\mathrm{gprod}\Big]. \quad (3.3)$$

The second form follows from Eqs. (2.5), (2.6) (check `Idef-P2`); in an era of matter $\mathcal{B} \approx a_{\rm sp}\,\mathrm{gsum}\,H_{\rm cf}/4$.

Second identity. Differentiate $\alpha_{\rm sy}$, Eq. (2.2), and use the conservation of momentum, $\mathcal{T}' = (a_{\rm sp}\,\mathrm{gsum} - 2H_{\rm cf})\,\mathcal{T} + k^2\,\mathcal{P} - k^2\mathcal{A}$:

$$2k^2\,\mathcal{Y}_S = \mathcal{Y}_T + \frac{3}{k^2}(\mathcal{Y}_M' + 2H_e\,\mathcal{Y}_M). \quad (3.4)$$

Both identities are verified by computer algebra for a mixture of pressureless matter, radiation with anisotropic stress, a fluid with a variable equation of state and a cosmological term (check `Idef-P3`), and numerically along the solutions of the program (Sec. 7.3).

3.2 The consequence

Without deflation $\mathcal{B} = 0$ and $\mathcal{Z} = 0$. Then (3.2) and (3.4) say that if one gap vanishes at all times, all three do: this is general relativity. With deflation $\mathcal{Z} \neq 0$, and **the three gaps cannot all vanish**. The equations of ordinary space with matter and radiation alone are inconsistent at the order gsum. The inconsistency is not a defect of the calculation: it is the statement that the agent of Sec. 1.1, which has a tension of the relative size gsum/H, takes part in the perturbations. The gaps are, up to the factor $8\pi G_N a_{\rm sp}^2$, the perturbations of its momentum density ($\mathcal{Y}_M$), of its pressure ($\mathcal{Y}_T$) and of its anisotropic stress ($\mathcal{Y}_S$). A gap of (E) would be a perturbation of its energy density, which part (ii) of A3 excludes.

The identities are two relations between three gaps. One more relation closes the system. Three natural choices set one gap to zero:

- **Closure "Momentum"**: $\mathcal{Y}_M = 0$. The agent carries no momentum density. Together with (ii), observers who move with the cold matter see neither an energy density nor a flow of energy of the agent: it is pure stress. Then

$$\mathcal{Y}_T = -\frac{\mathcal{Z}}{H_e}, \qquad \mathcal{Y}_S = -\frac{\mathcal{Z}}{2k^2H_e}.$$

  (E) and (M) are the two equations of the program of the main project, so that this closure changes that program least.
- **Closure "Pressure"**: $\mathcal{Y}_T = 0$. The agent carries no perturbation of its pressure: its density (zero) and its pressure are uniform on the surfaces of constant synchronous time, like the scale factors that it holds. Then

$$\mathcal{Y}_M = \mathcal{Z}, \qquad \mathcal{Y}_S = \frac{3}{2k^4}(\mathcal{Z}' + 2H_e\,\mathcal{Z}),$$

  and (M) is replaced by $2k^2\eta_{\rm sy}' = \mathcal{T} + \mathcal{Z}$.
- **Closure "Shear"**: $\mathcal{Y}_S = 0$. The agent carries no anisotropic stress, as a perfect fluid or a scalar field would not. Eliminating $\mathcal{Y}_T$ from (3.2) and (3.4) gives an equation of motion for the momentum density of the agent,

$$\mathcal{Y}_M' = -\Big(\frac{k^2}{3H_e} + 2H_e\Big)\mathcal{Y}_M + \frac{k^2}{3H_e}\,\mathcal{Z}. \quad (3.5)$$

  Far inside the horizon ($k \gg H_{\rm cf}$) the rate $k^2/3H_e$ is enormous, and $\mathcal{Y}_M$ relaxes at once to $\mathcal{Z}$, its value in the closure "Pressure": **inside the horizon the closure "Shear" is the closure**

“Pressure”, with corrections of the relative order $(H_{\rm cf}/k)^2$. Because of that rate Eq. (3.5) is stiff, and the integrator of the main project cannot follow it at an acceptable cost. The closure “Shear” is therefore not computed with the Einstein–Boltzmann program; it is integrated for pressureless matter with an implicit method in the independent script (check `Idef-P6`, Sec. 4).

All gaps are proportional to $a_{\rm sp}\,{\rm gsum}/H_{\rm cf} = {\rm gsum}/H$. For a rate of a few hundredths of H_0 that ratio is of the order 10^{-6} at recombination and of the order of the rate itself today (Sec. 7.3). The closure therefore cannot matter for anything that is decided at recombination, and must matter for the potentials and the growth of structure of the recent universe.

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4. The growth of pressureless matter

The clearest view of the closures is the density contrast δ_m of pressureless matter, for which $\mathcal{D} = \mathcal{E}_m\,\delta_m$ with $\mathcal{E}_m = 8\pi G_N a_{\rm sp}^2\rho_m$, $\mathcal{P} = \mathcal{A} = 0$, and $h'_{\rm sy} = -2\,\delta'_m$ (matter at rest in the synchronous frame). Each closure then gives one ordinary differential equation. For pressureless matter and a cosmological term it holds for every wavenumber; when radiation is present, far inside the horizon (check `Idef-P4`).

- *Closure “Pressure”.* (T) with (E) gives $h''_{\rm sy} + H_e\,h'_{\rm sy} + \mathcal{D} = 0$, that is

$$\delta''_m + H_e\,\delta'_m = \tfrac{1}{2}\,\mathcal{E}_m\,\delta_m \qquad\Longleftrightarrow\qquad \frac{d^2\delta_m}{dx4^2} + \Big(2H - \frac{\rm gsum}{2}\Big)\frac{d\delta_m}{dx4} = 4\pi\,G_N(x4)\,\rho_m\,\delta_m. \quad (4.1)$$

  Gravity is that of the moment, and the Hubble friction is reduced by ${\rm gsum}/2$.
- *Closure “Momentum”.* The derivative of (E), with (M) for $\eta'_{\rm sy}$, gives

$$\delta''_m + \frac{2H'_e + \mathcal{E}_m + \frac{4}{3}\mathcal{E}_r}{2H_e}\,\delta'_m = \frac{H_{\rm cf} - a_{\rm sp}\,{\rm gsum}}{2H_e}\,\mathcal{E}_m\,\delta_m. \quad (4.2)$$

  Here the *source* is reduced as well, by the factor $(H_{\rm cf} - a_{\rm sp}\,{\rm gsum})/(H_{\rm cf} - a_{\rm sp}\,{\rm gsum}/2)$. (The term $\frac{4}{3}\mathcal{E}_r$, with $\mathcal{E}_r = 8\pi G_N a_{\rm sp}^2\rho_r$ of the radiation, is the momentum density of radiation that streams freely: far inside the horizon such radiation has the velocity divergence $-h'_{\rm sy}/2$ (Ref. [1]), which enters the momentum constraint (M). Without the term Eq. (4.2) would not reduce to the familiar equation when the deflation is switched off.)
- *Closure “Shear”.* Far inside the horizon Eq. (4.1); in general no single equation.

Without deflation both reduce to the familiar $d^2\delta_m/dx4^2 + 2H\,d\delta_m/dx4 = 4\pi G_N\rho_m\delta_m$. *With* deflation that familiar equation, with $G_N(x4)$ inserted, is **neither** (4.1) nor (4.2). An earlier version of RS3 and of the manuscripts called it “the Newtonian limit of the eight-dimensional theory” and quoted its results; that was an error (RP9, item R-E33). For matter alone and three equal rates the three equations can be solved to the first order in ${\rm gsum}/H$: the growth rate $d\ln\delta_m/d\ln a_{\rm sp}$ is

$$1 - \tfrac{9}{16}\,\frac{\rm gsum}{H} \quad \text{(closure Pressure)}, \qquad 1 - \tfrac{13}{16}\,\frac{\rm gsum}{H} \quad \text{(closure Momentum)},$$
$$1 - \tfrac{11}{16}\,\frac{\rm gsum}{H} \quad \text{(the four-dimensional equation)}$$

(check `Idef-P4`; a unit test of the program reproduces the three coefficients to 10^{-4}).

For the model of RS3 (three equal rates, the Hubble constant that keeps the angle of the sound horizon, growth from $a_{\rm sp} = 0.01$ to today, relative to no deflation; command `matter`, check `def-D4`):

gsum/H_0	H_0	amplitude, M	amplitude, P	f_g, M	f_g, P	fixed H_0: amplitude, M	amplitude, P	four-dimensional equation
0	67.36	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
0.00100	67.29	1.0000	1.0002	1.0006	1.0011	0.9996	0.9998	0.9997
0.00324	67.15	1.0000	1.0008	1.0021	1.0035	0.9986	0.9993	0.9990
0.0100	66.70	1.0000	1.0024	1.0064	1.0109	0.9956	0.9979	0.9968
0.0300	65.36	1.0001	1.0073	1.0201	1.0339	0.9867	0.9937	0.9904
0.0644	62.98	1.0001	1.0160	1.0463	1.0773	0.9713	0.9863	0.9792
0.100	60.37	1.0001	1.0256	1.0780	1.1287	0.9553	0.9783	0.9674
0.300	41.94	1.0005	1.0907	1.4637	1.6751	0.8638	0.9288	0.8975

(M, P: the closures "Momentum" and "Pressure". Columns 3 to 6: ratios to the model without deflation at the Hubble constant of column 2, in km/s/Mpc, which keeps the angle. Columns 7 to 9: the ratio of the amplitudes at the fixed Hubble constant 67.36 km/s/Mpc, for the two closures and for the four-dimensional equation.)

1. *At a fixed Hubble constant* the amplitude of the density contrast today (the amplitude σ_8 for a given primordial amplitude) falls in both closures: $d\ln\sigma_8/d(\mathrm{gsum}/H_0) = -0.443$ ("Momentum") and -0.207 ("Pressure"). The four-dimensional equation gives -0.317, between the two.
2. *At a fixed angle of the sound horizon* the Hubble constant is lower (Sec. 6), the cosmological term takes over later, and matter clusters for longer. The net change is $+0.003$ ("Momentum") and $+0.239$ ("Pressure") per unit of gsum/H_0: $+0.001$ and $+0.077$ per cent at the rate that lunar laser ranging allows, $+0.01$ and $+1.60$ per cent at the rate that nucleosynthesis allows.
3. *The growth rate today* f_g rises in both closures: by the factors 1.0021 and 1.0035 (lunar laser ranging), 1.0463 and 1.0773 (nucleosynthesis).

Common to both closures is that, at a fixed angle, the amplitude does not fall and the growth rate rises. How large the changes are, and whether the amplitude changes at all, depends on the closure: **the growth of structure on the deflating spacetimes is not predicted by the hypotheses until the perturbations of the agent are specified.**

Verification. (a) An independent integration of the three equations reproduces the numbers of the program to 6.3×10^{-7} (check `Idef-P5`). (b) Integrated as systems for δ_m and the metric, with an implicit method and without using Eqs. (4.1), (4.2), the closures "Momentum" and "Pressure" give growth factors that do not depend on the wavenumber (to 7×10^{-14}), and the closure "Shear" differs from "Pressure" by 0.30 $(H_0/k)^2$: 1.5×10^{-6} at $k = 0.1$ Mpc^{-1} and 1.5×10^{-3} at $k = 0.003$ Mpc^{-1} (check `Idef-P6`). (c) The complete Einstein–Boltzmann calculation, with photons, neutrinos and baryons, reproduces the growth of Eqs. (4.1) and (4.2) to 1.3×10^{-4} (Sec. 7.5).

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5. The program

5.1 Copies of four modules, with a listed set of changes

The Einstein–Boltzmann program of the main project (S3, Sec. 2; provenance P8) has Newton's constant built into its units and the ordinary equation of the expansion built into its routines. The main project is paused and its program must not be changed. The revision therefore uses *copies* of the four modules that must change — the background and recombination, the perturbations, the spectra, the fits — and, so that a copy cannot drift from its original unnoticed, the copies are not edited by hand: the script `code/scripts/vendor_eb.py` reads the original files and applies a list of replacements, each of which must match the original text exactly once. Whatever is not on the list is the original code, character for character. The list is short:

1. *Expansion.* The Hubble rate is the positive root of Eq. (1.2); a table holds the clock $s = \mathrm{gsum}\,(x4 - x4_{\mathrm{today}})$, which obeys $ds/d\ln a_{\mathrm{sp}} = \mathrm{gsum}/H$, so that $G_N = G_N(\mathrm{today})\,e^s$.
2. *Helium.* At nucleosynthesis Newton's constant was e^s times today's value, with the value of s in the early universe. At a given temperature the square of the expansion rate is proportional to Newton's constant times the density of the radiation, which then consists of photons, electrons and positrons, and three neutrino species: 10.75 units of which one neutrino species is 7/4. It is therefore proportional to $e^s\,(\frac{43}{7} + \Delta N)$, where ΔN counts any radiation beyond the standard content that is present during nucleosynthesis, in units of one neutrino species (for the stepped fluids of the main project it is smaller than their density N after the step; S3, Sec. 3.4). For nucleosynthesis the smaller Newton's constant is thus equivalent to $(\frac{43}{7}+\Delta N)(e^s-1)$ additional neutrino species, a negative number, and the helium mass fraction Y_{He} is taken, with that number, from the same fit to the code PArthENoPE that the main project uses (Ref. [3]). This is the usual correspondence between the expansion rate and the number of species.
3. *Perturbations.* Every $8\pi G_N a_{\mathrm{sp}}^2 \rho$ is multiplied by e^s; (E) uses H_e; the closure "Pressure" replaces (M) by $2k^2\eta'_{\mathrm{sy}} = \mathcal{T} + \mathcal{Z}$.
4. *Sources of the temperature.* They contain α'_{sy} and α''_{sy} (S3, Sec. 2.1). The main project takes α'_{sy} from (S). With deflation
   (S) has a gap in either closure; the program adds the gap $\mathcal{Y}_S$ of the closure, an algebraic expression by Sec. 3.2, and for α''_{sy} the derivative of the gap along the solution.

Without deflation the copies are the original program. Every changed expression reduces to the original one operation for operation, and the spectra agree with those of the main project *to the last bit*: for the fiducial model, for superfluid dark matter and for a stepped dark fluid (unit tests of `spectra_defl.rs`; checks `def-E1`, `def-B1`, `def-F1`, `def-U1`). A second unit test runs the program with a rate of 10^{-30} in units of 100 km/s/Mpc, which is too small to change any number but makes the program take every branch of the deflating model; the potentials, sources and spectra are again those of the main project to the last bit.

5.2 Recombination on the deflating background

Recombination is computed as in the main project (the three-level atom with the corrections of the code RECFAST; S3, Sec. 2.1), on the expansion history of Eq. (1.2) and with the helium fraction of item 2. For the three models of the command `background` of the main project (check `def-B2`):

model	gsum/H_0	held	H_0	z_*	r_s (Mpc)	$100\,\theta_*$	Y_{He}	G_N ratio
fiducial	0	H_0	67.36	1089.92	144.45	1.03965	0.2454	1.0000
fiducial	0.00324	H_0	67.36	1089.86	144.68	1.04028	0.2451	0.9969
fiducial	0.00324	angle	67.15	1089.86	144.68	1.03965	0.2451	0.9969
fiducial	0.0644	H_0	67.36	1088.76	149.09	1.05252	0.2402	0.9400
fiducial	0.0644	angle	62.98	1088.72	149.23	1.03965	0.2401	0.9383
fluid, $N = 0.5$	0	H_0	67.36	1090.40	142.02	1.02248	0.2520	1.0000
fluid, $N = 0.5$	0.00324	H_0	67.36	1090.34	142.25	1.02310	0.2517	0.9969
fluid, $N = 0.5$	0.00324	angle	67.15	1090.34	142.25	1.02248	0.2517	0.9969
fluid, $N = 0.5$	0.0644	H_0	67.36	1089.22	146.59	1.03517	0.2468	0.9400
fluid, $N = 0.5$	0.0644	angle	62.97	1089.19	146.73	1.02248	0.2466	0.9383
stepped, $N = 0.5$	0	H_0	67.36	1090.25	142.55	1.02627	0.2481	1.0000
stepped, $N = 0.5$	0.00324	H_0	67.36	1090.19	142.77	1.02689	0.2479	0.9969
stepped, $N = 0.5$	0.00324	angle	67.15	1090.19	142.77	1.02627	0.2479	0.9969
stepped, $N = 0.5$	0.0644	H_0	67.36	1089.07	147.13	1.03900	0.2429	0.9400
stepped, $N = 0.5$	0.0644	angle	62.97	1089.04	147.27	1.02627	0.2428	0.9383

(The models with extra radiation are the fluid without step and the stepped fluid of the model with

$z_t = 2 \times 10^4$, each with $N = 0.5$. Column "held": the row keeps either the Hubble constant of the main project or the angle θ_* of the same model without deflation. H_0 in km/s/Mpc; r_s: sound horizon at last scattering; "G_N ratio": Newton's constant at last scattering divided by its value today. The data file contains in addition the drag epoch, its sound horizon, the redshift of reionization and the age.) Weaker gravity in the past means slower expansion at a given density. The sound horizon is larger, the angle θ_* larger, and the helium fraction lower. Last scattering is slightly later: the redshift z_* falls, because in a universe that expands more slowly a photon spends more time, and meets more electrons, in each interval of redshift, so that the optical depth reaches one at a lower redshift. The two models with dark radiation respond exactly as the fiducial model does.

Independent verification. A separate calculation of recombination with Peebles' three-level atom, written in Python and sharing no code with the program (check `Idef-C2`), gives $z_* = 1090.84$ without deflation against 1089.92, as expected of the simpler atomic model, and reproduces the *shifts* caused by the deflation: -0.551 against -0.550 for gsum $= 0.03\,H_0$ and -1.882 against -1.877 for $0.1\,H_0$.

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6. The Hubble constant at a fixed angle

RS3, Sec. 2, infers the Hubble constant by keeping the angle θ_* of the sound horizon at last scattering. Its first version held the redshift of last scattering and the helium fraction at their values without deflation. With the thermal history of Sec. 5.2 neither is held (command `cosmology`, check `def-C2`; table in RS3, Sec. 2). The slope at small rates is

$$\frac{d \ln H_0}{d(\text{gsum}/H_0)} = -0.969 \qquad (\text{with } z_* \text{ and } Y_{\text{He}} \text{ held fixed: } -0.911),$$

and $H_0 = 66.70$, 65.36, 60.37 km/s/Mpc for gsum $= 0.01$, 0.03, $0.1\,H_0$ (held fixed: 66.74, 65.48, 60.77). **Holding z_* and Y_{He} fixed underestimates the fall of the Hubble constant by 5.8 to 6.0 per cent of itself.** Most of the difference is the shift of last scattering: at a later last scattering the sound horizon is larger still. The helium fraction accounts for 0.045 km/s/Mpc of it at the rate that nucleosynthesis allows (check `def-E7`), less than a fifth. The independent calculation of Sec. 5.2 confirms both sets of numbers: 65.365 and 60.367 km/s/Mpc in the full calculation (program: 65.365, 60.368), 65.483 and 60.773 with last scattering and helium held fixed (65.483, 60.773). At the bounds of RS3, Sec. 3, the inferred Hubble constant changes by -0.211 km/s/Mpc (lunar laser ranging) and -4.38 km/s/Mpc (nucleosynthesis). The approximation is recorded as item R-E34 of the provenance RP9.

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7. The spectra at a fixed angle

7.1 The models

Five rates are computed, each with three equal rates of deflation, with the densities of baryons and of cold dark matter, the primordial amplitude and tilt and the optical depth of the fiducial model of the main project, and with the Hubble constant that keeps $100\,\theta_* = 1.03965$ (command `spectrum`, check `def-E2`):

gsum/H_0	H_0	G_N ratio	Y_{He}	σ_8, M	σ_8, P	largest change	ideal	$\Delta\chi^2$
0	67.360	1.0000	0.2454	0.8228	0.8228	0	0.0	0.00
0.00324	67.149	0.9969	0.2451	0.8219	0.8225	2.7×10^{-3}	3.1	1.12
0.0100	66.704	0.9905	0.2446	0.8200	0.8219	8.3×10^{-3}	9.7	10.79

gsum/H_0	H_0	G_N ratio	Y_{He}	σ_8, M	σ_8, P	largest change	ideal	$\Delta\chi^2$
0.0300	65.365	0.9714	0.2430	0.8142	0.8201	0.025	29.5	98.85
0.0644	62.976	0.9383	0.2401	0.8032	0.8160	0.056	64.8	468.30
0.100	60.368	0.9037	0.2370	0.7903	0.8104	0.090	103.5	1166.70

(The rates $0.00324\,H_0$ and $0.0644\,H_0$ are the largest that lunar laser ranging and nucleosynthesis allow, RS3, Sec. 3. H_0 in km/s/Mpc. "G_N ratio": Newton's constant at last scattering divided by its value today. M, P: the closures "Momentum" and "Pressure". "Largest change": of D_ℓ^{TT} relative to the model without deflation for $30 \le \ell \le 2500$. "Ideal": the root of the sum over those multipoles of the squared change divided by the variance $2/(2\ell+1)$ of a single multipole, that is, the number of standard deviations by which a measurement of the temperature on the whole sky without noise would tell the model from no deflation if nothing else were unknown. $\Delta\chi^2$: against the Planck-like mock data of the main project with all other parameters fixed. The last three columns are for the closure "Momentum". The redshift of last scattering and the sound horizon of these models are in the table of RS3, Sec. 2.)

7.2 What remains when the angle is kept

Keeping θ_* keeps the acoustic peaks in place: the first five move by at most 0.32 in ℓ.

peak	ℓ: no deflation	D_ℓ^{TT}	ℓ: lunar ranging rate	D_ℓ^{TT}	ℓ: nucleosynthesis rate	D_ℓ^{TT}
1	220.9	5740	220.9	5741	220.9	5752
2	537.4	2607	537.4	2608	537.5	2623
3	815.2	2584	815.2	2585	815.3	2612
4	1131.4	1264	1131.4	1265	1131.5	1287
5	1429.3	844	1429.3	845	1429.5	865

(Positions and heights, in $\mu\mathrm{K}^2$, of the first five maxima of D_ℓ^{TT}.) What changes is their height, by at most 0.12 per cent at the rate of lunar laser ranging and 2.5 per cent at the rate of nucleosynthesis, and the damping tail (check `def-E3`). The reasons are the same as for any change of the expansion rate at recombination: a slower expansion changes the ratio of the diffusion length to the sound horizon, the width of the last-scattering surface, and the decay of the potentials shortly after recombination; and the lower helium fraction leaves more electrons per baryon.

- *At the rate that lunar laser ranging allows*, the temperature spectrum differs from that of the model without deflation by at most 2.7×10^{-3} of itself, 0.134 of the cosmic variance of a single multipole; $\Delta\chi^2 = 1.12$, less than the 3.84 of a deviation of two standard deviations. Data of the quality of Planck cannot tell this rate from no deflation.
- *At the rate that nucleosynthesis allows* Newton's constant at last scattering was 0.9383 of today's, the spectrum changes by up to 5.6 per cent, and $\Delta\chi^2 = 468$ before any parameter is refitted. The microwave background is then a stronger test than nucleosynthesis (Sec. 8).

The helium fraction matters at this level. At the rate of nucleosynthesis the slower expansion of the early universe is equivalent to -0.379 neutrino species, and the helium fraction falls from 0.24540 to 0.24008. Ignoring this would change the temperature spectrum by up to 8.6×10^{-3} of itself and $\Delta\chi^2$ from 468.3 to 335.3 (check `def-E7`).

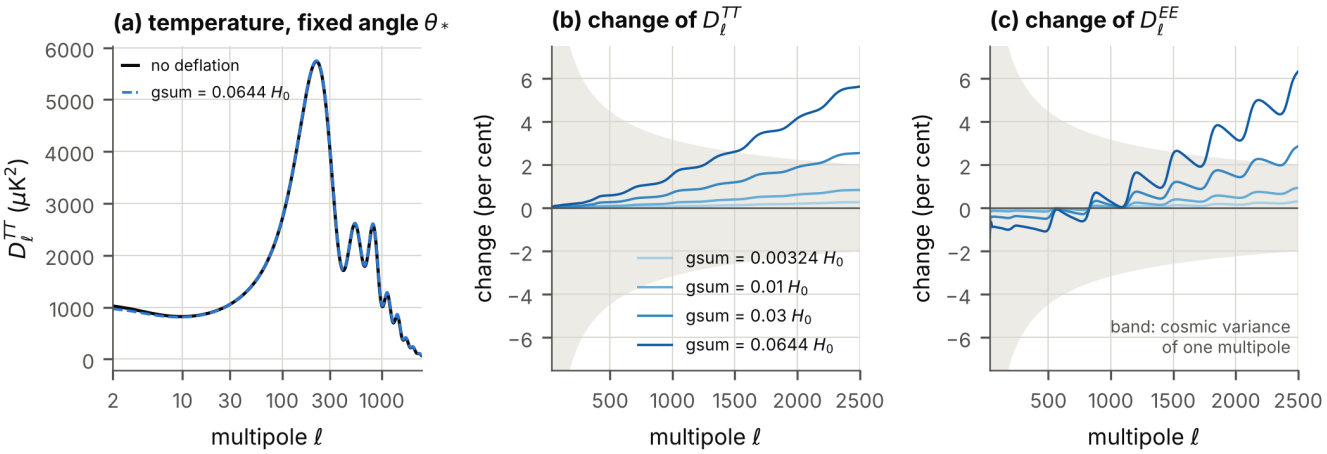


Figure 1: The spectra of the microwave background at a fixed angle of the sound horizon. (a) The temperature spectrum without deflation and at the largest rate that nucleosynthesis allows. (b), (c) The change of the temperature and polarization spectra relative to the model without deflation for four rates (darker: faster), with the cosmic variance of a single multipole as a band. Closure "Momentum"; at these multipoles the closure "Pressure" gives the same curves. Data: `data/cmb_deflation_cl.csv`, `data/cmb_deflation_summary.csv`; program: `deflation spectrum`.

7.3 The closure

gsum/H_0	$\ell \geq 30$: difference	share of the effect	$\ell < 30$: difference	share of the effect	σ_8 ratio, M	σ_8 ratio, P
0.00324	1.4×10^{-3}	1.4×10^{-3}	0.011	2.14	0.9989	0.9997
0.0100	4.3×10^{-3}	1.4×10^{-3}	0.037	2.15	0.9966	0.9990
0.0300	0.012	1.3×10^{-3}	0.129	2.16	0.9896	0.9967
0.0644	0.023	1.2×10^{-3}	0.330	2.13	0.9762	0.9918
0.100	0.031	9.9×10^{-4}	0.556	2.04	0.9605	0.9850

("Difference": the largest difference between the temperature spectra of the two closures in the range of multipoles, in units of the cosmic variance of a single multipole. "Share of the effect": the root of the sum of squares of that difference divided by the root of the sum of squares of the effect of the deflation itself, in the same range. σ_8 ratio: to the model without deflation; M, P: the closures "Momentum" and "Pressure".) Check `def-E4`:

- **For $30 \leq \ell \leq 2500$ the closure does not matter.** The spectra of the two closures differ by a thousandth of the effect of the deflation (1.4×10^{-3} and 1.2×10^{-3} of it at the two bounds), and $\Delta\chi^2$ against the mock data is the same to five digits. Everything in Secs. 7.2 and 8 follows from A1, A2 and the first two parts of A3, whatever the closure.
- **For $\ell < 30$ it does.** There the potentials of the recent universe act (the late integrated Sachs–Wolfe effect), and the closures differ by more than the effect of the deflation in either of them (2.1 times). The largest angular scales are not a prediction of the hypotheses. Their cosmic variance is large; at the rate of nucleosynthesis the closures differ by at most a third of it.
- **The amplitude of matter** σ_8 is 0.9762 of its value without deflation in the closure "Momentum" and 0.9918 in the closure "Pressure" at the rate of nucleosynthesis (0.99892 and 0.99969 at the rate of lunar laser ranging).

The table that follows shows, for the wavenumber $k = 0.02$ Mpc^{-1} and five times, what the agent must supply in each closure (M: "Momentum", P: "Pressure"), and the numerical checks of the identities of Sec. 3.1 (check `def-E5`). The column gsum/H is the ratio a_{sp} gsum/H_{cf}. The gaps are divided by that ratio times the size of the terms of their equations — $k^2 H_{\mathrm{cf}} |\mathcal{U}|$ for (M), $(k^2 + H_{\mathrm{cf}}^2)|\mathcal{U}|$

for (T), $|\mathcal{U}|$ for (S) — so that a number of order one means "of the first order in the rate". The data file has the same for $k = 0.005$ and 0.08 Mpc^{-1}.

closure	$\tau_{\rm cf}$ (Mpc)	gsum$/H$	$\mathcal{U}$	y_M	y_T	y_S	check of (3.4)	check of (3.2)
M	280	2.7×10^{-6}	0.7373	0.000	-0.507	-0.2679	3×10^{-5}	5×10^{-5}
M	800	3.0×10^{-5}	0.5402	0.000	-0.656	-0.3314	4×10^{-5}	7×10^{-5}
M	3000	9.4×10^{-4}	0.5178	0.000	-0.666	-0.3333	4×10^{-4}	9×10^{-4}
M	9000	0.021	0.5083	0.000	-0.708	-0.3543	4×10^{-4}	9×10^{-4}
M	13000	0.055	0.4633	0.000	-0.980	-0.4900	4×10^{-4}	7×10^{-4}
P	280	2.7×10^{-6}	0.7373	0.536	0.000	0.1355	3×10^{-5}	5×10^{-5}
P	800	3.0×10^{-5}	0.5402	0.663	0.000	0.0289	4×10^{-5}	7×10^{-5}
P	3000	9.4×10^{-4}	0.5181	0.666	0.000	0.0028	4×10^{-4}	9×10^{-4}
P	9000	0.021	0.5139	0.697	0.000	0.0004	4×10^{-4}	9×10^{-4}
P	13000	0.055	0.4810	0.936	0.000	0.0004	4×10^{-4}	7×10^{-4}

For the three wavenumbers of the data file together, and after recombination (the last four times), the closure "Momentum" lacks 0.54 to 0.98 in the trace equation and 0.31 to 0.49 in the equation of the shear, in these units; in the closure "Pressure" the momentum constraint lacks 0.62 to 0.94, and the equation of the shear lacks 2.94 to 4.84 times $(H_{\rm cf}/k)^2$ — far inside the horizon ($k > 20\,H_{\rm cf}$) between 2.5×10^{-5} and 7.0×10^{-3}: there the closure "Pressure" satisfies the equation of the shear as well, as Eq. (3.5) says. The two identities hold along the numerical solutions to 8.6×10^{-4}, which is the accuracy of the numerical derivative with which they are tested. The ratio $a_{\rm sp}\,\text{gsum}/H_{\rm cf} = \text{gsum}/H$ is 2.7×10^{-6} at recombination, 5.5×10^{-2} at the last time of the table and 6.9×10^{-2} today.

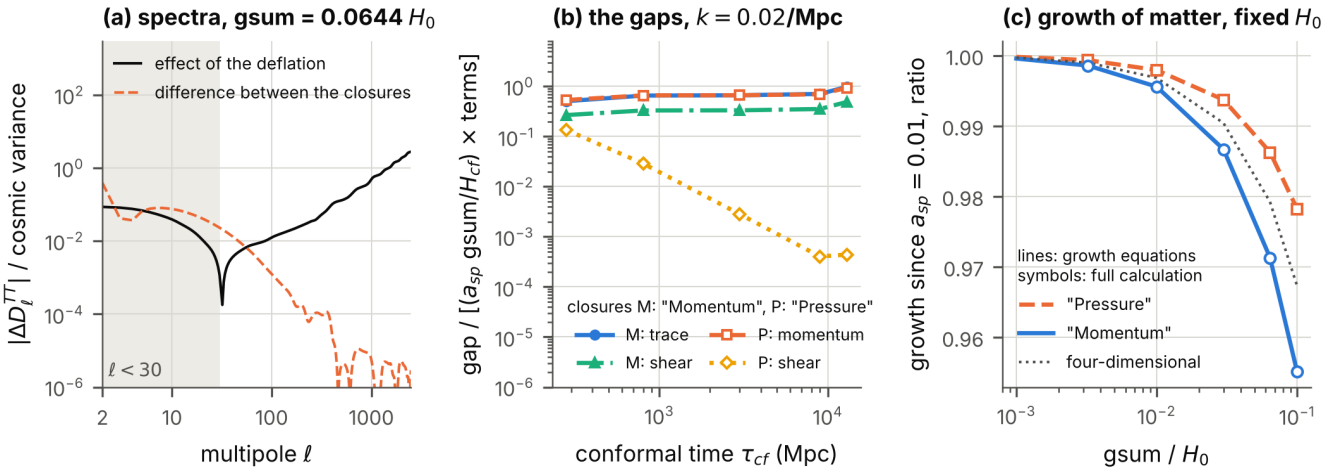


Figure 2: The closure of the perturbation equations at the largest rate that nucleosynthesis allows. (a) The difference between the temperature spectra of the closures "Momentum" and "Pressure", and the effect of the deflation itself, in units of the cosmic variance of a single multipole: the closure matters below $\ell = 30$ only. (b) What the agent supplies: the gaps of the momentum constraint, the trace equation and the equation of the shear, in units of $a_{\rm sp}\,\text{gsum}/H_{\rm cf}$ times the terms of each equation. (c) The growth of the density contrast of matter since $a_{\rm sp} = 0.01$ relative to no deflation: the growth equations (4.1), (4.2) (lines), the complete calculation at $k = 0.1$ Mpc^{-1} (symbols), and the four-dimensional equation, which is neither closure. Data: `data/cmb_deflation_closure.csv`, `data/cmb_deflation_gaps.csv`, `data/cmb_deflation_growth.csv`, `data/growth_deflation.csv`.

7.4 Numerical convergence

Halving all tolerances and all wavenumber steps changes the spectra of the deflating model with gsum $= 0.03\,H_0$, in either closure, by at most 1.5×10^{-4} (TT), 2.0×10^{-4} (TE, in units of the geometric mean of TT and EE) and 3.6×10^{-4} (EE) for $\ell \geq 100$, and by 1.2×10^{-3}, 3.3×10^{-3}, 3.5×10^{-3} for $30 \leq \ell < 100$; σ_8 changes by 2.5×10^{-7} (check `def-E6`). These are the numbers of the main project for its fiducial model (1.4×10^{-4}, 1.8×10^{-4}, 3.4×10^{-4}; 1.2×10^{-3}, 4.1×10^{-3}, 5.9×10^{-3}), and its requirements are met.

7.5 The growth of structure in the complete calculation

At the fixed Hubble constant 67.36 km/s/Mpc:

gsum$/H_0$	program, M	Eq. (4.2)	program, P	Eq. (4.1)
0	1.00000	1.00000	1.00000	1.00000
0.00324	0.99856	0.99857	0.99933	0.99933
0.0100	0.99556	0.99557	0.99791	0.99792
0.0300	0.98664	0.98668	0.99368	0.99371
0.0644	0.97123	0.97131	0.98620	0.98627
0.100	0.95513	0.95525	0.97819	0.97830

At the Hubble constant that keeps the angle:

gsum$/H_0$	H_0	growth, M	growth, P	σ_8 ratio, M	σ_8 ratio, P
0	67.360	1.00000	1.00000	1.00000	1.00000
0.00324	67.149	1.00001	1.00077	0.99892	0.99969
0.0100	66.704	1.00001	1.00239	0.99663	0.99901
0.0300	65.365	1.00003	1.00727	0.98955	0.99672
0.0644	62.976	1.00004	1.01597	0.97620	0.99177
0.100	60.368	1.00001	1.02545	0.96051	0.98497

(Growth of the density contrast of cold dark matter and baryons at $k = 0.1$ Mpc^{-1} from $a_{\rm sp} = 0.01$ to today, relative to no deflation. “Program”: the complete Einstein–Boltzmann calculation in the closure “Momentum” (M) or “Pressure” (P); the columns headed by an equation: the growth equation of that closure. H_0 in km/s/Mpc.) The complete calculation and the growth equation of the same closure agree to 1.3×10^{-4} for the five rates, at a fixed Hubble constant and at a fixed angle; with the equation of the *other* closure they would disagree by at least $0.24\,\text{gsum}/H_0$ (check `def-E8`; independently `Idef-P7`). The amplitude σ_8 contains, besides this growth, the change of the scale $8\,h^{-1}$ Mpc with the Hubble constant and the growth before $a_{\rm sp} = 0.01$ under a weaker gravity.

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8. What mock data say about the rate

8.1 The question, and the method

S3, Sec. 3.3, asks how far the Hubble constant can be raised by dark radiation before the fit to microwave-background data degrades, and answers with mock data: the spectra of the fiducial model stand in for the measurements (so that the fiducial model fits with $\Delta\chi^2 = 0$), with the noise and sky coverage of the Planck satellite, and of Planck together with a deep survey of a quarter of

the sky. The same question is put here to the deflation: *given such data, how large can* gsum *be, and what happens to the Hubble constant?* The likelihood, the two data sets, the six parameters (ω_b, ω_c, h, the amplitude A_s and tilt n_s of the primordial spectrum, the optical depth τ) and the fitting procedure are those of the main project, used from its library unchanged (check `def-F1`: its Fisher errors are reproduced to 3.1×10^{-11}). The rate is the seventh parameter. Because the rates are positive by hypothesis, the derivative of the spectra with respect to gsum at zero rate is one-sided.

8.2 The Fisher matrix

mock data	closure	error of the rate	bound	six held	slope	error of H_0	rate free
Planck-like	M	0.01963	0.0385	0.00108	-84.35	0.561	1.748
Planck-like	P	0.01974	0.0387	0.00108	-84.40	0.561	1.758
with deep survey	M	0.01056	0.0207	0.00055	-82.11	0.372	0.943
with deep survey	P	0.01058	0.0207	0.00055	-82.13	0.372	0.945

("Error of the rate": the standard error of gsum$/H_0$ with the six parameters free. "Bound": 1.96 times that error. "Six held": the error if the six parameters are held fixed. "Slope": the change of the best-fit Hubble constant in km/s/Mpc per unit of gsum$/H_0$. Last two columns: the standard error of the Hubble constant in km/s/Mpc with the rate held at zero, and with the rate free. M, P: the closures. Checks `def-F2`, `def-F3`.) With the six parameters free, Planck-like data determine gsum$/H_0$ to 0.01963 (one standard deviation), and the upper bound at 95 per cent — 1.96 standard deviations, for a parameter that cannot be negative and whose best value is zero — is

$$\begin{aligned} &\text{gsum} < 0.0385\, H_0 = 2.65 \times 10^{-12} \text{ per year} && \text{(Planck-like)},\\ &\text{gsum} < 0.0207\, H_0 && \text{(with the deep survey)}. \end{aligned}$$

With the six parameters *held* the error would be 0.00108: nearly all of the effect of the deflation on the spectra — above all the shift of the angle θ_* — can be absorbed by the other parameters, and only one part in eighteen of the sensitivity survives. The closure changes the error by half a per cent (0.01963 against 0.01974).

The bound lies between those of RS3: it is 11.9 times weaker than that of lunar laser ranging ($0.00324\, H_0$) and 1.7 times stronger than that of nucleosynthesis ($0.0644\, H_0$). Unlike lunar laser ranging it tests the rate at early times, and unlike both it is a forecast for mock data, not a measurement.

The Hubble constant falls along the degeneracy: by 84.35 km/s/Mpc per unit of gsum$/H_0$ for the Planck-like data and 82.11 with the deep survey, that is by 3.2 and 1.7 km/s/Mpc at the respective bounds. (At a fixed angle and fixed densities the slope is 65.24, Sec. 6; in the fit the density of dark matter rises and the tilt falls as well.) The error of the Hubble constant grows from 0.561 to 1.748 km/s/Mpc when the rate is free — entirely towards smaller values, since the rate cannot be negative.

8.3 Fits at fixed rate

mock data	closure	gsum$/H_0$	$\Delta\chi^2$	H_0	ω_c	n_s	σ_8	G_N ratio
Planck-like	M	0.00324	0.027	67.086	0.12009	0.9638	0.8218	0.9969
Planck-like	M	0.00500	0.065	66.937	0.12014	0.9633	0.8213	0.9952
Planck-like	M	0.0100	0.261	66.512	0.12029	0.9616	0.8198	0.9905
Planck-like	P	0.0100	0.258	66.512	0.12029	0.9616	0.8218	0.9905
Planck-like	M	0.0150	0.589	66.084	0.12043	0.9599	0.8183	0.9857

mock data	closure	gsum/H_0	$\Delta\chi^2$	H_0	ω_c	n_s	σ_8	G_N ratio
Planck-like	M	0.0200	1.051	65.654	0.12058	0.9583	0.8168	0.9809
Planck-like	M	0.0300	2.398	64.784	0.12087	0.9549	0.8136	0.9713
Planck-like	M	0.0644	11.575	61.695	0.12193	0.9430	0.8016	0.9379
Planck-like	P	0.0644	11.467	61.695	0.12193	0.9430	0.8145	0.9379
with deep survey	M	0.00324	0.094	67.094	0.12008	0.9637	0.8218	0.9969
with deep survey	M	0.00500	0.225	66.948	0.12013	0.9631	0.8212	0.9952
with deep survey	M	0.0100	0.904	66.534	0.12025	0.9613	0.8197	0.9905
with deep survey	P	0.0100	0.900	66.534	0.12025	0.9613	0.8216	0.9905
with deep survey	M	0.0150	2.040	66.117	0.12038	0.9594	0.8181	0.9857
with deep survey	M	0.0200	3.644	65.699	0.12050	0.9576	0.8165	0.9809
with deep survey	M	0.0300	8.301	64.851	0.12076	0.9539	0.8131	0.9713
with deep survey	M	0.0644	39.791	61.842	0.12169	0.9408	0.8007	0.9380
with deep survey	P	0.0644	39.655	61.841	0.12170	0.9408	0.8136	0.9380

(Check `def-F4`. Each row is a fit of the six parameters at the given rate; H_0 in km/s/Mpc; "G_N ratio": Newton's constant at last scattering divided by its value today. The data file contains all parameters of each fit.) For rates up to $0.01\,H_0$ the minimum $\Delta\chi^2$ is 1.00 to 1.01 times the value $(\text{gsum}/\text{error})^2$ of the Fisher matrix, and the shift of the Hubble constant 1.00 to 1.01 times its slope: the Fisher matrix describes the fits. The profile reaches $\Delta\chi^2 = 3.84$ at gsum $= 0.0376\,H_0$ (Planck-like) and $0.0205\,H_0$ (with the deep survey), against the bounds 0.0385 and 0.0207 of the Fisher matrix. At the rate of lunar laser ranging the best fit has $\Delta\chi^2 = 0.027$ and $H_0 = 67.086$ km/s/Mpc. At the rate of nucleosynthesis it has $\Delta\chi^2 = 11.6$ (3.4 standard deviations) and $H_0 = 61.70$ km/s/Mpc for the Planck-like data, and $\Delta\chi^2 = 39.8$ (6.3 standard deviations) with the deep survey: **the largest rate that nucleosynthesis allows would be excluded by data of the quality of Planck.**

Accuracy (check `def-F5`). Recomputing the derivatives of the spectra at the best fit for gsum $= 0.02\,H_0$ and continuing the fit changes $\Delta\chi^2$ by less than 0.001 and the Hubble constant by 0.001 km/s/Mpc. Refitting with the closure "Pressure" changes $\Delta\chi^2$ by at most 0.9 per cent and the Hubble constant by 0.0005 km/s/Mpc.

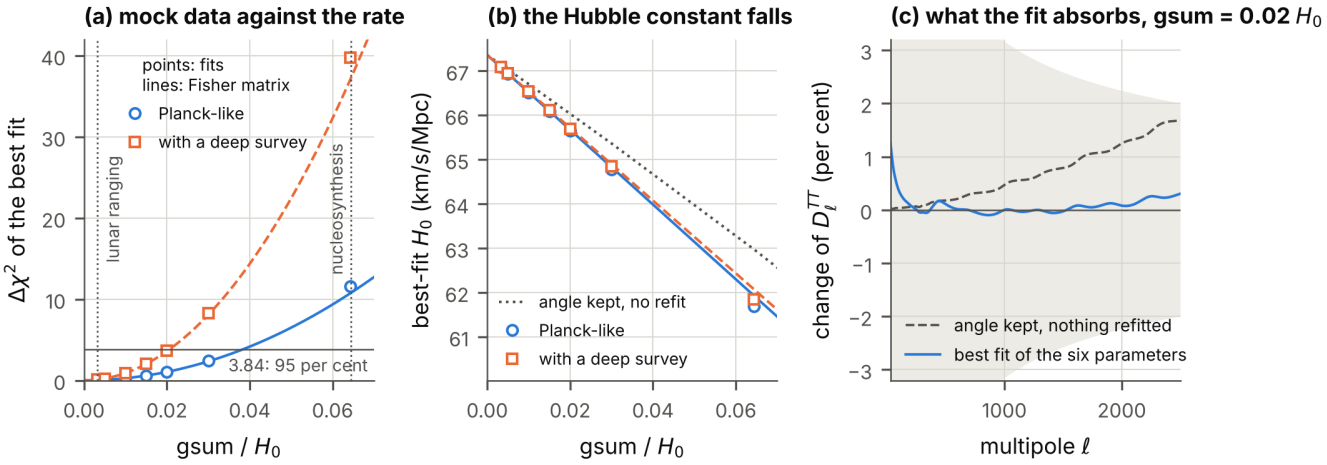


Figure 3: Mock data of the microwave background against the rate of deflation. (a) The minimum $\Delta\chi^2$ of fits of the six parameters at fixed rate (symbols) and the parabola of the Fisher matrix (lines), for the two data sets; the rates that lunar laser ranging and nucleosynthesis allow are marked. (b) The best-fit Hubble constant, with the slope of the Fisher matrix and, dotted, the Hubble constant that keeps the angle when nothing else is refitted. (c) The temperature spectrum at gsum $= 0.02\,H_0$ relative to the data: with the angle kept and nothing refitted, and at the best fit, which absorbs most of the difference; the band is the cosmic variance of a single multipole. Data: `data/deflation_profile.csv`, `data/deflation_fisher.csv`, `data/deflation_residuals.csv`, `data/hubble_deflation.csv`; program: `deflation forecast`.

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9. The dark-radiation families of the main project on the deflating background

S3, Sec. 3, fits seven one-parameter families of extra radiation of the density N (in units of one neutrino species) to the mock data: free-streaming radiation, a fluid, the stepped fluid of the model at four transition redshifts z_t, and the Wess–Zumino stepped fluid. (The tables below use the names of S3: "model fluid" is the stepped fluid of the model, "Wess–Zumino fluid" that of the Wess–Zumino model.) The command `families` repeats the complete calculation — Fisher matrices, fits at eight values of N for both data sets, the check of the derivatives, the scan of the transition redshift — with three equal rates of deflation at the largest total rate that lunar laser ranging allows, gsum $= 0.00324\,H_0$. The data remain the spectra of the fiducial model without deflation, so that $\Delta\chi^2$ is counted from a model that has neither deflation nor extra radiation. A second part, Sec. 9.3, treats the rate and the density N as free parameters together.

9.1 The Fisher matrix at the rate of lunar laser ranging

The deflating model *without* extra radiation fits the Planck-like data with $\Delta\chi^2 = 0.027$ at $H_0 = 67.086$ km/s/Mpc, and the data with the deep survey with $\Delta\chi^2 = 0.094$ at $H_0 = 67.094$ km/s/Mpc (check `def-N1`; the first pair is the fit of Sec. 8.3). The derivatives of the spectra with respect to the six parameters and to N are taken at that best fit, on the deflating background.

mock data	family	error of N: no deflation	deflating	dH_0/dN: no deflation	deflating
Planck-like	free-streaming radiation	0.1431	0.1432	8.190	8.180
Planck-like	fluid, no step	0.1352	0.1352	10.441	10.435

mock data	family	error of N: no deflation	deflating	dH_0/dN: no deflation	deflating
Planck-like	model fluid, $z_t = 10^4$	0.1222	0.1222	5.631	5.624
Planck-like	model fluid, $z_t = 2 \times 10^4$	0.0846	0.0846	4.684	4.676
Planck-like	model fluid, $z_t = 5 \times 10^4$	0.0602	0.0602	9.283	9.273
Planck-like	Wess–Zumino fluid, $z_t = 2 \times 10^4$	0.1601	0.1598	8.578	8.573
Planck-like	model fluid, $z_t = 10^6$	0.1500	0.1500	10.042	10.038
with deep survey	free-streaming radiation	0.0770	0.0771	7.923	7.915
with deep survey	fluid, no step	0.0667	0.0666	9.801	9.797
with deep survey	model fluid, $z_t = 10^4$	0.0858	0.0858	6.275	6.271
with deep survey	model fluid, $z_t = 2 \times 10^4$	0.0591	0.0591	5.399	5.395
with deep survey	model fluid, $z_t = 5 \times 10^4$	0.0421	0.0421	6.928	6.919
with deep survey	Wess–Zumino fluid, $z_t = 2 \times 10^4$	0.0879	0.0876	8.246	8.241
with deep survey	model fluid, $z_t = 10^6$	0.0742	0.0741	9.445	9.442

("Error of N": one standard deviation with the six parameters free. dH_0/dN: the change of the best-fit Hubble constant in km/s/Mpc per unit of N. "No deflation": the main project, file `data/dr_fisher.csv`.) The errors of N agree with those of the main project to 0.2 per cent for the Planck-like data and to 0.3 per cent with the deep survey: at this rate the deflation does not change what the data say about extra radiation (check `def-N2`).

9.2 Fits at a fixed density of extra radiation

mock data	family	N	H_0: no deflation	deflating	shift	$\Delta\chi^2$: no deflation	deflating
Planck-like	no extra radiation	0	67.360	67.086	-0.274	0	0.027
Planck-like	free-streaming radiation	0.2	68.996	68.722	-0.274	1.92	1.57
Planck-like	free-streaming radiation	0.5	71.455	71.179	-0.276	11.55	10.67
Planck-like	free-streaming radiation	1.0	75.533	75.255	-0.278	43.83	42.10
Planck-like	fluid, no step	0.2	69.445	69.171	-0.274	2.17	1.79
Planck-like	fluid, no step	0.5	72.548	72.273	-0.275	13.32	12.34
Planck-like	fluid, no step	1.0	77.746	77.469	-0.277	51.94	50.01
Planck-like	model fluid, $z_t = 10^4$	0.2	68.485	68.211	-0.274	2.64	2.38
Planck-like	model fluid, $z_t = 10^4$	0.5	70.174	69.898	-0.276	16.08	15.40
Planck-like	model fluid, $z_t = 10^4$	1.0	72.991	72.714	-0.277	61.23	59.89
Planck-like	model fluid, $z_t = 2 \times 10^4$	0.2	68.292	68.019	-0.274	5.47	5.21
Planck-like	model fluid, $z_t = 2 \times 10^4$	0.5	69.698	69.422	-0.276	32.98	32.32
Planck-like	model fluid, $z_t = 2 \times 10^4$	1.0	72.025	71.747	-0.278	124.24	122.93
Planck-like	model fluid, $z_t = 5 \times 10^4$	0.2	69.216	68.942	-0.274	10.81	10.18
Planck-like	model fluid, $z_t = 5 \times 10^4$	0.5	71.968	71.695	-0.274	65.29	63.71
Planck-like	model fluid, $z_t = 5 \times 10^4$	1.0	76.625	76.351	-0.275	247.48	244.38
Planck-like	Wess–Zumino fluid, $z_t = 2 \times 10^4$	0.2	69.072	68.798	-0.274	1.55	1.22
Planck-like	Wess–Zumino fluid, $z_t = 2 \times 10^4$	0.5	71.641	71.365	-0.276	9.38	8.52
Planck-like	Wess–Zumino fluid, $z_t = 2 \times 10^4$	1.0	75.926	75.647	-0.279	35.77	34.11
Planck-like	model fluid, $z_t = 10^6$	0.2	69.364	69.091	-0.274	1.76	1.44
Planck-like	model fluid, $z_t = 10^6$	0.5	72.345	72.071	-0.275	10.80	9.97

mock data	family	N	H_0: no deflation	deflating	shift	$\Delta\chi^2$: no deflation	deflating
Planck-like	model fluid, $z_t = 10^6$	1.0	77.334	77.058	-0.277	42.16	40.49
with deep survey	no extra radiation	0	67.360	67.094	-0.266	0	0.094
with deep survey	free-streaming radiation	0.2	68.943	68.676	-0.267	6.63	5.36
with deep survey	free-streaming radiation	0.5	71.311	71.044	-0.268	40.18	36.94
with deep survey	free-streaming radiation	1.0	75.230	74.961	-0.269	153.44	147.06
with deep survey	fluid, no step	0.2	69.315	69.048	-0.267	8.89	7.29
with deep survey	fluid, no step	0.5	72.193	71.926	-0.267	54.51	50.44
with deep survey	fluid, no step	1.0	77.000	76.731	-0.269	211.85	203.79
with deep survey	model fluid, $z_t = 10^4$	0.2	68.614	68.347	-0.267	5.37	4.70
with deep survey	model fluid, $z_t = 10^4$	0.5	70.487	70.219	-0.268	32.60	30.85
with deep survey	model fluid, $z_t = 10^4$	1.0	73.616	73.346	-0.269	124.07	120.58
with deep survey	model fluid, $z_t = 2 \times 10^4$	0.2	68.436	68.169	-0.267	11.22	10.72
with deep survey	model fluid, $z_t = 2 \times 10^4$	0.5	70.035	69.766	-0.269	67.34	66.03
with deep survey	model fluid, $z_t = 2 \times 10^4$	1.0	72.694	72.423	-0.270	252.39	249.73
with deep survey	model fluid, $z_t = 5 \times 10^4$	0.2	68.742	68.475	-0.267	22.10	20.77
with deep survey	model fluid, $z_t = 5 \times 10^4$	0.5	70.759	70.493	-0.266	132.70	129.29
with deep survey	model fluid, $z_t = 5 \times 10^4$	1.0	74.148	73.879	-0.268	498.87	492.12
with deep survey	Wess–Zumino fluid, $z_t = 2 \times 10^4$	0.2	69.005	68.738	-0.267	5.17	4.00
with deep survey	Wess–Zumino fluid, $z_t = 2 \times 10^4$	0.5	71.456	71.189	-0.267	31.49	28.50
with deep survey	Wess–Zumino fluid, $z_t = 2 \times 10^4$	1.0	75.542	75.274	-0.268	121.43	115.53
with deep survey	model fluid, $z_t = 10^6$	0.2	69.243	68.976	-0.267	7.18	5.79
with deep survey	model fluid, $z_t = 10^6$	0.5	72.016	71.749	-0.267	44.03	40.45
with deep survey	model fluid, $z_t = 10^6$	1.0	76.641	76.373	-0.268	171.23	164.11

(Each row is a fit of the six parameters at the fixed density N; H_0 in km/s/Mpc; "shift": deflating minus not deflating. The data file has eight values of N for each family.) Three things can be read off (check `def-N2`).

1. *Every Hubble constant is lower by about a quarter of a km/s/Mpc.* For all seven families and all eight densities the best-fit Hubble constant on the deflating background lies 0.274 to 0.279 km/s/Mpc below that of the main project for the Planck-like data, and 0.266 to 0.270 km/s/Mpc below it with the deep survey. The model without extra radiation is lower by 0.274 and 0.266 km/s/Mpc: this is what the slopes of Sec. 8.2 give for this rate. The effects of the deflation and of the extra radiation on the Hubble constant add.
2. *The fits are slightly better than without deflation.* The deflating model without extra radiation fits worse than the model that produced the data. Yet for every family and every density $N \geq 0.1$ the minimum $\Delta\chi^2$ on the deflating background is *lower* than in the main project, by up to 3.11 at $N = 1$ for the Planck-like data and by up to 8.06 with the deep survey. The reason is given in Sec. 9.3.
3. *The measures of the main project.* S3, Sec. 3.4, summarizes each family by the density at which the best fit reaches $\Delta\chi^2 = 4$, by the Hubble constant there, and by the $\Delta\chi^2$ at which the best-fit Hubble constant reaches the local measurement of 73.50 km/s/Mpc. On both backgrounds:

family	N at $\Delta\chi^2 = 4$: no deflation	deflating	H_0 there: no deflation	deflating	$\Delta\chi^2$ at $H_0 = 73.50$: no deflation	deflating
free-streaming radiation	0.29	0.31	69.74	69.62	25	26

family	N at $\Delta\chi^2 = 4$: no deflation	deflating	H_0 there: no deflation	deflating	$\Delta\chi^2$ at $H_0 = 73.50$: no deflation	deflating
fluid, no step	0.27	0.29	70.20	70.12	19	19
model fluid, $z_t = 10^4$	0.25	0.26	68.75	68.53	72*	77*
model fluid, $z_t = 2 \times 10^4$	0.17	0.18	68.16	67.90	211*	228*
model fluid, $z_t = 5 \times 10^4$	0.12	0.13	68.48	68.27	113	121
Wess–Zumino fluid, $z_t = 2 \times 10^4$	0.32	0.35	70.13	70.06	19	19
model fluid, $z_t = 10^6$	0.30	0.32	70.39	70.32	16	17

family	N at $\Delta\chi^2 = 4$: no deflation	deflating	H_0 there: no deflation	deflating	$\Delta\chi^2$ at $H_0 = 73.50$: no deflation	deflating
free-streaming radiation	0.16	0.18	68.59	68.48	95	98
fluid, no step	0.13	0.15	68.67	68.59	87	90
model fluid, $z_t = 10^4$	0.17	0.19	68.44	68.25	120	126*
model fluid, $z_t = 2 \times 10^4$	0.12	0.12	68.00	67.75	330*	356*
model fluid, $z_t = 5 \times 10^4$	0.08	0.09	67.94	67.70	412	441
Wess–Zumino fluid, $z_t = 2 \times 10^4$	0.18	0.20	68.81	68.74	69	71
model fluid, $z_t = 10^6$	0.15	0.17	68.76	68.69	76	78

(First table: Planck-like mock data; second table: with the deep survey. The interpolation is that of the main project: between the fitted densities the square root of $\Delta\chi^2$ and the Hubble constant are linear in N; an asterisk marks a linear extrapolation beyond $N = 1$. H_0 in km/s/Mpc.) Because the fits are slightly better, slightly more extra radiation is allowed at $\Delta\chi^2 = 4$; the Hubble constant reached there is nevertheless lower than in the main project, and the local value costs the same or more. For free-streaming radiation and the Planck-like data: $N = 0.31$ instead of 0.29, $H_0 = 69.62$ instead of 69.74 km/s/Mpc, and $\Delta\chi^2 = 26$ instead of 25 at the local value. **At the rate that lunar laser ranging allows, the deflation changes none of the conclusions of the main project about its dark radiation, and what it changes goes against a larger Hubble constant.**

9.3 The rate and the extra radiation as free parameters together

Why are the fits with extra radiation better on the deflating background? The answer is in the Fisher matrix with eight parameters — the six of the standard model, the density N and the rate — at the fiducial model (check `def-N4`):

family	error of N, rate zero	error of N, rate free	error of the rate, N free	correlation	N per unit rate	net slope	largest rise of H_0
free-streaming radiation	0.1431	0.2505	0.0344	0.821	5.98	-35.3	2.30
fluid, no step	0.1352	0.2490	0.0362	0.840	5.79	-23.9	2.77
model fluid, $z_t = 10^4$	0.1222	0.1451	0.0233	0.540	3.36	-65.4	1.35
model fluid, $z_t = 2 \times 10^4$	0.0846	0.0911	0.0211	0.370	1.60	-76.9	0.78
model fluid, $z_t = 5 \times 10^4$	0.0602	0.0760	0.0248	0.610	1.87	-67.0	1.10
Wess–Zumino fluid, $z_t = 2 \times 10^4$	0.1601	0.3503	0.0429	0.889	7.26	-22.1	2.69
model fluid, $z_t = 10^6$	0.1500	0.2496	0.0326	0.799	6.11	-23.0	2.95

family	error of N, rate zero	error of N, rate free	error of the rate, N free	correlation	N per unit rate	net slope	largest rise of H_0
free-streaming radiation	0.0770	0.1540	0.0211	0.866	6.32	-32.1	1.20
fluid, no step	0.0667	0.1763	0.0279	0.926	5.85	-24.8	1.28
model fluid, $z_t = 10^4$	0.0858	0.1012	0.0125	0.530	4.31	-55.0	1.06
model fluid, $z_t = 2 \times 10^4$	0.0591	0.0617	0.0110	0.283	1.58	-73.6	0.63
model fluid, $z_t = 5 \times 10^4$	0.0421	0.0485	0.0121	0.495	1.97	-68.4	0.57
Wess–Zumino fluid, $z_t = 2 \times 10^4$	0.0879	0.2121	0.0255	0.910	7.57	-19.7	1.42
model fluid, $z_t = 10^6$	0.0742	0.1767	0.0251	0.908	6.38	-21.9	1.37

(First table: Planck-like mock data; second table: with the deep survey. Errors: one standard deviation of N and of gsum$/H_0$, with the other of the two held at zero or free; the six parameters are free throughout. "Correlation": the correlation coefficient of the two. "N per unit rate": the best-fitting density at a fixed rate, divided by gsum$/H_0$; it is the correlation coefficient times the error of N divided by the error of the rate, both with both free. "Net slope": the change of the best-fit Hubble constant, in km/s/Mpc per unit of gsum$/H_0$, when N follows its best-fitting value: the slope of Sec. 8.2 plus dH_0/dN times the preceding column. "Largest rise of H_0": the largest increase of the best-fit Hubble constant, in km/s/Mpc, among all models with $N \geq 0$, gsum ≥ 0 and $\Delta\chi^2 \leq 3.84$; in every row it is reached at zero rate.)

1. *Consistency.* With the rate held at zero the errors of N are those of the main project, to 2.6×10^{-11}.
2. *The density of extra radiation and the rate are positively correlated* for every family: the correlation coefficient is 0.37 to 0.89 for the Planck-like data and 0.28 to 0.93 with the deep survey. Extra radiation makes the early universe expand faster; the weaker gravity of the deflating spacetime makes it expand more slowly; in the spectra the two compensate in part. With both free, the error of gsum$/H_0$ from the Planck-like data grows from 0.0196 to between 0.0211 and 0.0429, depending on the family, and the error of N grows by the same factor.
3. *This is why the fits of Sec. 9.2 are better.* For two correlated parameters $\Delta\chi^2$ contains a term proportional to the product of N and the rate, with a negative sign. At a fixed positive rate it lowers $\Delta\chi^2$ in proportion to N, and the best-fitting density is no longer zero: at the rate that lunar laser ranging allows it is $N = 0.005$ to 0.023, with $\Delta\chi^2 = 0.006$ to 0.023 instead of 0.027 (Planck-like). The fits confirm the matrix: at $N = 0.1$ and 0.2 the differences between the minimum $\Delta\chi^2$ of Sec. 9.2 and that of the main project are 0.99 to 1.01 times the term of the Fisher matrix.
4. *The compensation does not help the Hubble constant.* When the rate grows and N follows its best-fitting value, extra radiation raises the Hubble constant and the deflation lowers it. The sum, the column "net slope", is negative for every family and both data sets: -35.3 km/s/Mpc per unit of gsum$/H_0$ for free-streaming radiation and the Planck-like data, against -84.35 without extra radiation. The largest Hubble constant with $\Delta\chi^2 \leq 3.84$ is therefore reached at *zero* rate in all fourteen cases: a rise of 0.78 to 2.95 km/s/Mpc for the Planck-like data. **The deflation lowers the Hubble constant faster than it makes room for extra radiation; it cannot help the dark radiation of the model, or any of the other families, to raise the Hubble constant.**

9.4 Accuracy

Recomputing the derivatives of the spectra at the best fit for $N = 0.5$ of each family and continuing the fit changes $\Delta\chi^2$ by at most 0.054 and the Hubble constant by at most 0.050 km/s/Mpc. The scan of the transition redshift of the stepped fluid of the model at $N = 0.4$, from $z_t = 10^3$ to 10^6, has the shape that it has in the main project, with Hubble constants lower by 0.272 to 0.277 km/s/Mpc (check `def-N3`; files `data/dr_deflation_chord_check.csv`, `data/dr_deflation_zt_scan.csv`). The threshold 3.84 is that of a deviation of 1.96 standard deviations in one parameter, here the Hubble

constant; Sec. 9.3 rests on the Fisher matrix, which Sec. 9.2 confirms for densities up to 0.2 and at the rate of lunar laser ranging only.

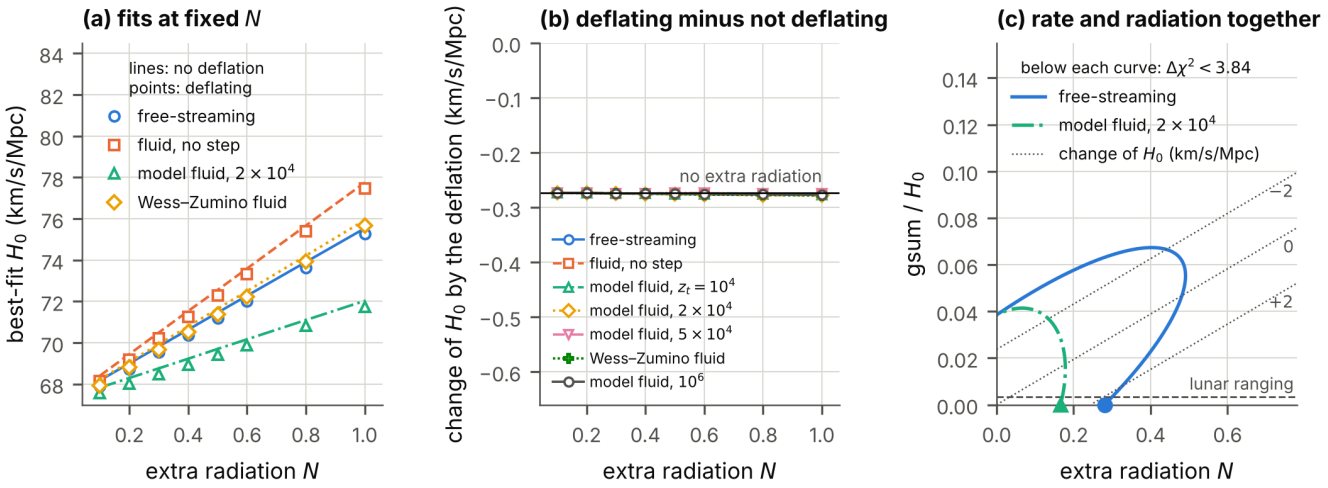


Figure 4: The dark-radiation families of the main project on the deflating background. (a) The best-fit Hubble constant against the density of extra radiation for four families: main project (lines) and deflating background at the rate that lunar laser ranging allows (symbols). (b) The difference for all seven families; the horizontal line is the shift of the model without extra radiation. (c) The rate and the extra radiation as free parameters together: the regions with $\Delta\chi^2 < 3.84$ of the Fisher matrix with eight parameters for two families, lines of constant change of the best-fit Hubble constant for free-streaming radiation (dotted), the point of the largest Hubble constant of each region (filled symbols, at zero rate), and the rate that lunar laser ranging allows (dashed). Planck-like mock data. Data: `data/dr_deflation_profiles.csv`, `data/dr_deflation_joint.csv`, and `data/dr_profiles.csv` of the main project; program: `deflation families`.

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10. Superfluid dark matter on the deflating background

S3, Secs. 4.1 to 4.3, asks what becomes of the growth of structure if the dark matter is the superfluid itself, a fluid with first sound, in two limits: a degenerate gas of fermions of mass m with eight internal states (weak pairing) and a condensate of bound pairs with the sound speed c_{s0} today. Pressure suppresses structure below a wavenumber; the *half mode* is the wavenumber at which the power falls to a quarter of that of cold dark matter, and the bound of the Lyman-α forest on thermal warm dark matter, translated into a half mode, limits m from below and c_{s0} from above. The command `superfluid` repeats the calculation on four backgrounds: no deflation; the rate of lunar laser ranging; the rate of nucleosynthesis in each of the two closures. Each background has the Hubble constant that keeps the angle.

dark matter	no deflation	lunar ranging rate	nucleosynthesis rate, M	nucleosynthesis rate, P	change (per cent)
fermions, $m = 0.3$ keV	6.215	6.206	6.025	6.025	-3.07
fermions, $m = 0.5$ keV	11.222	11.205	10.875	10.875	-3.09
fermions, $m = 0.75$ keV	17.996	17.969	17.444	17.444	-3.07
fermions, $m = 1$ keV	25.196	25.159	24.426	24.426	-3.06
fermions, $m = 1.5$ keV	40.657	40.590	39.379	39.379	-3.14
fermions, $m = 2$ keV	57.249	57.159	55.401	55.401	-3.23
fermions, $m = 3$ keV	92.774	92.633	89.868	89.868	-3.13

dark matter	no deflation	lunar ranging rate	nucleosynthesis rate, M	nucleosynthesis rate, P	change (per cent)
fermions, $m = 4.5$ keV	150.645	150.419	146.013	146.010	-3.08
condensate, $c_{s0}/c = 3.0 \times 10^{-12}$	125.047	124.857	121.132	121.132	-3.13
condensate, $c_{s0}/c = 1.0 \times 10^{-11}$	56.870	56.781	55.003	55.003	-3.28
condensate, $c_{s0}/c = 3.0 \times 10^{-11}$	27.777	27.736	26.922	26.922	-3.08
condensate, $c_{s0}/c = 1.0 \times 10^{-10}$	12.734	12.714	12.333	12.333	-3.15
condensate, $c_{s0}/c = 1.0 \times 10^{-9}$	2.911	2.906	2.822	2.822	-3.04
condensate, $c_{s0}/c = 1.0 \times 10^{-8}$	0.693	0.692	0.670	0.670	-3.28
condensate, $c_{s0}/c = 4.5 \times 10^{-8}$	0.280	0.280	0.271	0.271	-3.25

deflation	relic (keV)	half mode (Mpc^{-1})	fermion mass (keV)	largest c_{s0}/c	in m/s	R_{TF} (pc)
none	3.5	25.94	1.0249	3.335×10^{-11}	0.01000	0.740
none	5.3	41.11	1.5140	1.645×10^{-11}	0.00493	0.365
lunar ranging rate	3.5	25.94	1.0262	3.327×10^{-11}	0.00997	0.739
lunar ranging rate	5.3	41.11	1.5161	1.641×10^{-11}	0.00492	0.364
nucleosynthesis rate, M	3.5	25.94	1.0523	3.178×10^{-11}	0.00953	0.705
nucleosynthesis rate, M	5.3	41.11	1.5554	1.565×10^{-11}	0.00469	0.347
nucleosynthesis rate, P	3.5	25.94	1.0523	3.178×10^{-11}	0.00953	0.705
nucleosynthesis rate, P	5.3	41.11	1.5554	1.565×10^{-11}	0.00469	0.347

(First table: the half-mode wavenumber, in Mpc^{-1}, of fifteen models on the four backgrounds; M, P: the closures "Momentum" and "Pressure"; "change": of the column "nucleosynthesis rate, M" relative to no deflation. Second table: the limits that follow when the half mode is required not to lie below that of a thermal relic of 3.5 or 5.3 keV (column "relic"), the bounds of the Lyman-α forest that the main project uses. "Half mode": that of the relic. "Fermion mass": the smallest allowed mass m. The largest allowed sound speed of a condensate today is given in units of the speed of light and in m/s. R_{TF} is the radius of the core that a condensate with that sound speed would form, with Newton's constant of today.)

1. *Without deflation* the copy of the program reproduces the two files of the main project to 3.4×10^{-11} (check `def-U1`).
2. *The suppression moves to slightly smaller wavenumbers.* The half-mode wavenumber falls by 0.15 to 0.16 per cent at the rate that lunar laser ranging allows and by 3.0 to 3.3 per cent at the rate that nucleosynthesis allows, for all fifteen models (check `def-U2`). The reason is the weaker gravity of the time at which the suppression is set, long before today. The Jeans wavenumber, below which pressure wins over gravity, is proportional to the square root of Newton's constant, and in the early universe Newton's constant was 0.9969 and 0.9383 of its present value at the two rates: the square roots differ from one by -0.154 and -3.13 per cent.
3. *The limits move accordingly.* For the benchmark of 5.3 keV the smallest fermion mass moves from 1.514 keV without deflation to 1.516 keV at the rate of lunar laser ranging and 1.555 keV at the rate of nucleosynthesis; the largest sound speed of a condensate from $1.645 \times 10^{-11}\, c$ to $1.641 \times 10^{-11}\, c$ and $1.565 \times 10^{-11}\, c$.
4. *The closure does not matter:* the two closures give the same half-mode wavenumbers to 1.5×10^{-5}. The suppression is set long before the closure can act (Sec. 3.2).

Two cautions. (a) The benchmark wavenumbers are those of the standard model, in which the bounds of the Lyman-α forest were derived. At the rate of nucleosynthesis the expansion rate at the redshifts of the forest differs from that of the standard model by a few per cent as well, and with it the relation between wavenumbers and the velocities that the forest measures. The last rows

of the second table therefore show the size of the effect; they are not limits of the same standing as those of the main project. At the rate that lunar laser ranging allows both changes are below two tenths of a per cent. (b) The equation of state of the dark fluid in ordinary space is that of the main project. For the degenerate gas it follows from the density and the mass of the fermions alone. For the condensate it contains the strength of a contact interaction. Whether that strength, seen from ordinary space, would itself follow the volume of the deflating coordinates, as the couplings of RS1 do, depends on the microscopic model; if it grew at the rate gsum, the sound speed at a given density would have been smaller in the past, which acts against the shift found here. The calculation does not decide this.

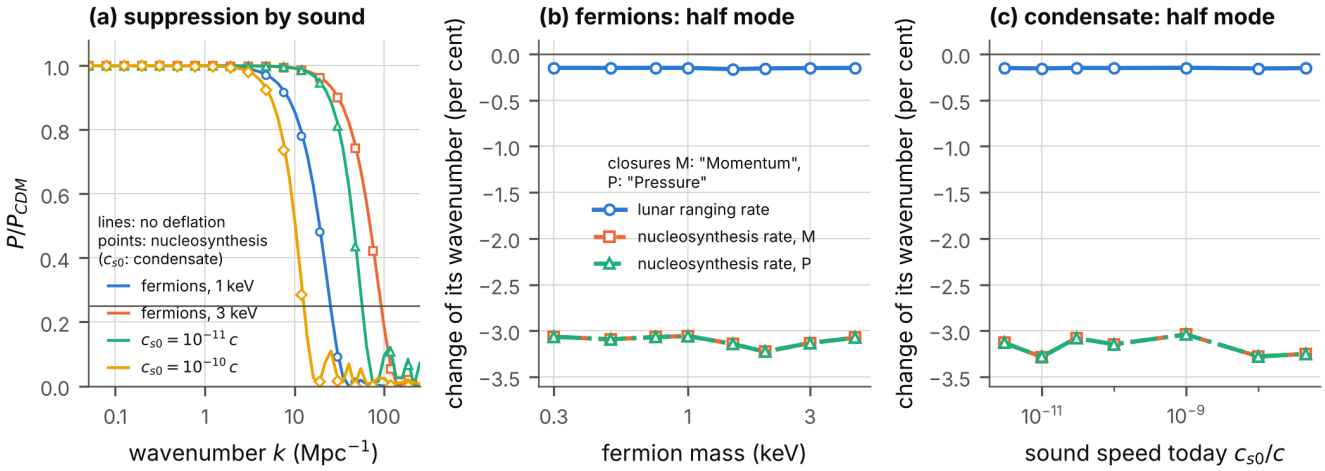


Figure 5: Superfluid dark matter on the deflating background. (a) The power spectrum of matter relative to cold dark matter for two fermion masses and two sound speeds of a condensate: no deflation (lines) and the largest rate that nucleosynthesis allows (symbols). (b), (c) The change of the half-mode wavenumber for the fermions and for the condensate on the three deflating backgrounds. Data: `data/sfdm_deflation_transfer.csv`, `data/sfdm_deflation_summary.csv`; program: `deflation superfluid`.

---

11. What has been established, and what has not

Established, each under the assumptions named:

1. the Einstein equations of ordinary space to first order with unperturbed extra scale factors, Eqs. (2.3), (2.4), for arbitrary rates and both signatures (assumption A3, part (i); computer algebra);
2. that these equations are inconsistent with matter alone at the order gsum, the two identities (3.2), (3.4) between the gaps, and the three closures; that the closure “Shear” coincides with “Pressure” inside the horizon (A1, A2, A3);
3. the growth equations (4.1), (4.2), and that the four-dimensional equation with $G_N(x4)$ is neither;
4. that the copies of the Einstein–Boltzmann program reproduce the main project to the last bit without deflation;
5. the Hubble constant at a fixed angle with recombination on the deflating background, $d\ln H_0/d(\mathrm{gsum}/H_0) = -0.969$ (A1, A2), confirmed by an independent calculation of recombination;
6. the spectra for $\ell \geq 30$ at five rates, their independence of the closure, and the forecast gsum $< 0.0385\,H_0$ (Planck-like), $0.0207\,H_0$ (with a deep survey) (A1, A2, parts (i) and (ii) of A3);
7. that the Hubble constant falls in every fit;
8. for the seven dark-radiation families of the main project, at the rate that lunar laser ranging

allows: best-fit Hubble constants lower by about a quarter of a km/s/Mpc, and slightly better fits (A1, A2, parts (i) and (ii) of A3; fits with the complete calculation);

9. that the density of extra radiation and the rate of deflation are positively correlated, and that the largest Hubble constant with $\Delta\chi^2 \leq 3.84$ is reached at zero rate for every family (Fisher matrix, confirmed by the fits at the rate of lunar laser ranging);
10. the shift of the scale of suppression of superfluid dark matter, and its independence of the closure (A1, A2, A3; with the two cautions of Sec. 10).

Not established.

1. *The agent.* Nothing in the hypotheses says what keeps the rates constant, and therefore nothing says which closure holds, or whether any of the three does. The growth of structure, the amplitude σ_8 and the spectra below $\ell = 30$ are computed for two closures and differ at the first order in gsum/H_0. They are the quantities that a theory of the agent would have to supply.
2. *Part (i) of A3.* That the extra scale factors do not respond to inhomogeneities is an assumption. In a theory in which they are dynamical fields, they would be perturbed, and the equations of Sec. 2 would contain additional terms.
3. *The theory of gravity.* Everything rests on A2. If gravity is of the Einstein–Gauss–Bonnet kind in which the deflating spacetimes with equal rates are vacuum solutions (DMD, Sec. 1.4a), Eq. (1.1) and with it this whole supplement would have to be derived anew (RS3, Sec. 9).
4. *Real data.* Sections 8 and 9 are forecasts for mock data without gravitational lensing. No measured spectrum has been fitted.
5. *A varying rate.* The rates are constant throughout cosmic history in every calculation.
6. *Nonlinear structure.* Only linear perturbations are computed.

Appendix A. Complete output of the programs

A.1 `deflation background`

```
PROJECT spacetimes, Revision_Deflation: checks of the thermal history on the deflating spacetime
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation background
2 of 2 checks passed

[PASS] def-B1  (exact) thermal history without deflation: for the three models of the command
 ↪  `background` of the main project (fiducial; fluid dark radiation; stepped fluid) the copy of
 ↪  the background and recombination modules gives the numbers of the main project to the last
 ↪  bit: conformal age, redshifts of last scattering, of the maximum of the visibility and of the
 ↪  drag epoch, the two sound horizons, the two angles, the redshift of reionization and the
 ↪  helium fraction
[PASS] def-B2  (numerical) thermal history on the deflating spacetime (assumptions A1, A2;
 ↪  data/background_deflation.csv). Gravity was weaker in the past. At the rate that
 ↪  nucleosynthesis allows (gsum = 0.06436 H_0) and the Hubble constant of the main project,
 ↪  Newton's constant at last scattering is 0.9400 of today's, the sound horizon at last
 ↪  scattering grows from 144.449 to 149.092 Mpc, 100 theta_* from 1.03965 to 1.05252, last
 ↪  scattering moves from z_* = 1089.92 to 1088.76, the helium fraction falls from 0.24540 to
 ↪  0.24023, and the age grows from 13.814 to 13.955 Gyr; the same directions hold for the two
 ↪  models with dark radiation and at the rate of lunar laser ranging. The Hubble constant that
 ↪  restores the angle is 62.976 km/s/Mpc for the fiducial model, 62.970 with fluid dark radiation
 ↪  (N = 0.5) and 62.972 with the stepped fluid of the model (against 67.360, 67.360, 67.360
 ↪  without deflation at the same angles)
[PASS] ALL  (numerical) every check above passed
```

A.2 deflation spectrum

```
PROJECT spacetimes, Revision_Deflation: checks of the spectra of the microwave background on the
 ↪  deflating spacetime
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation spectrum
8 of 8 checks passed

[PASS] def-E1  (exact) the Einstein-Boltzmann modules of the revision (copies of four modules of
 ↪  the main project with a listed set of changes, written by code/scripts/vendor_eb.py) reproduce
 ↪  the program of the main project when the rates of deflation are zero, for both closures: the
 ↪  spectra TT, TE and EE for 2 <= l <= 2500, the power spectrum of matter at 164 wavenumbers,
 ↪  sigma_8, the angle theta_* and the sound horizon agree bit for bit; and the spectra agree with
 ↪  the file data/cmb_fiducial_cl.csv of the main project to the digits written there (largest
 ↪  relative deviation 4.9e-11)
[PASS] def-E2  (numerical) spectra at a fixed angle of the sound horizon (assumptions A1, A2, A3;
 ↪  table data/cmb_deflation_summary.csv). With the full calculation of recombination on the
 ↪  deflating background the Hubble constant that keeps 100 theta_* = 1.03965 (kept to 5e-15) is
 ↪  67.149, 66.704, 65.365, 62.976, 60.368 km/s/Mpc for gsum = 0.00324, 0.01, 0.03, 0.06436, 0.1
 ↪  H_0 (H_0 = 67.36 km/s/Mpc): it falls as the rate grows. Last scattering moves from z_* =
 ↪  1089.92 to 1088.04 at the largest rate, and the helium fraction falls from 0.24540 to 0.23696.
 ↪  If z_* and the helium fraction are held at their values without deflation, the fall of the
 ↪  Hubble constant is underestimated by 5.8 to 6.0 per cent of itself. The acoustic peaks stay
 ↪  where they are (the first five move by at most 0.32 in l); their heights change by at most
 ↪  0.12 per cent at the rate of lunar laser ranging and 2.5 per cent at the rate of
 ↪  nucleosynthesis (data/cmb_deflation_peaks.csv)
[PASS] def-E3  (numerical) what remains in the spectra when the angle is kept. At the rate that
 ↪  lunar laser ranging allows (gsum = 0.00324 H_0) the temperature spectrum differs from that of
 ↪  the model without deflation by at most 2.7e-3 of itself for 30 <= l <= 2500 (0.134 of the
 ↪  cosmic variance of a single multipole); chi^2 against Planck-like mock data with all other
 ↪  parameters fixed is 1.12 (closure "Momentum") and 1.12 ("Pressure"), less than the 3.84 of a
 ↪  deviation of two standard deviations: such data cannot tell this rate from no deflation. At
 ↪  the rate that nucleosynthesis allows (gsum = 0.06436 H_0, Newton's constant at last scattering
 ↪  0.9383 of today's) the difference reaches 0.056 of the spectrum and chi^2 = 468: the microwave
 ↪  background is then the stronger test (command `forecast`)
[PASS] def-E4  (numerical) dependence on the closure of the perturbation equations ("Momentum":
 ↪  the agent that holds the extra scale factors carries no momentum density; "Pressure": it
 ↪  carries no pressure perturbation; file data/cmb_deflation_closure.csv). For 30 <= l <= 2500,
 ↪  the multipoles that the forecast uses, the temperature spectra of the two closures differ by
 ↪  at most 1.4e-3 and 2.3e-2 of the cosmic variance of a multipole at gsum = 0.00324 and 0.06436
 ↪  H_0, which is 1.4e-3 and 1.2e-3 of the effect of the deflation itself (root of the sum of
 ↪  squares); chi^2 against Planck-like mock data is 1.122 against 1.122, and 468.30 against
 ↪  468.30. The spectra at these multipoles are a prediction of assumptions A1 and A2 alone. For l
 ↪  < 30, where the potentials of today act (late integrated Sachs-Wolfe effect), the closures
 ↪  differ by up to 0.011 and 0.330 of the cosmic variance, 2.14 and 2.13 of the effect; and the
 ↪  amplitude sigma_8 of matter is 0.9762 of its value without deflation in the closure "Momentum"
 ↪  against 0.9918 in the closure "Pressure" at gsum = 0.06436 H_0 (0.99892 against 0.99969 at
 ↪  0.00324 H_0). There the result depends on the closure and is not a prediction of the hypotheses
[PASS] def-E5  (numerical) what the agent supplies (file data/cmb_deflation_gaps.csv; gsum =
 ↪  0.06436 H_0, three wavenumbers, five times). Each closure is a consistent theory: the quantity
 ↪  eta' + alpha'' that the source of the temperature uses equals the derivative of the two
 ↪  potentials along the solution (largest relative difference 1.7e-4), and the two identities
 ↪  from which the program computes the gaps hold along the solution to 8.6e-4, the accuracy of a
 ↪  numerical derivative. After recombination the gaps are of the first order in the rate: in
 ↪  units of (a gsum/Hc) times the terms of their equations, the closure "Momentum" lacks 0.31 to
 ↪  0.49 in the equation of the shear and 0.54 to 0.98 in the trace equation; the closure
 ↪  "Pressure" lacks 0.62 to 0.94 in the momentum constraint, and in the equation of the shear
 ↪  2.94 to 4.84 times (Hc/k)^2 (far inside the horizon, k > 20 Hc: 2.5e-5 to 7.0e-3): there the
 ↪  closure "Pressure" satisfies the equation of the shear as well, and is the closure "Shear".
 ↪  The ratio a gsum/Hc = gsum/H is 2.7e-6 at the first of the five times (recombination), 5.5e-2
 ↪  at the last (conformal time 13000 Mpc) and 6.9e-2 today: the closure matters only for what
 ↪  happens late
```

```
[PASS] def-E6  (numerical) numerical convergence on the deflating spacetime (gsum = 0.03 H_0, both
 ↪  closures; file data/cmb_deflation_convergence.csv): halving all tolerances and wavenumber
 ↪  steps changes the spectra by at most 1.5e-4 (TT), 2.0e-4 (TE, in units of sqrt(TT EE)), 3.6e-4
 ↪  (EE) for 100 <= l <= 2500, and by 1.2e-3, 3.3e-3, 3.5e-3 for 30 <= l <= 99; sigma_8 changes by
 ↪  2.5e-7. The program of the main project has 1.4e-4, 1.8e-4, 3.4e-4 and 1.2e-3, 4.1e-3, 5.9e-3
 ↪  for its fiducial model (its file data/cmb_convergence.csv); the requirements are those of the
 ↪  main project (5e-4 for l >= 100; 2e-3 for TT and 1e-2 for TE and EE below)
[PASS] def-E7  (numerical) the helium fraction. Newton's constant at nucleosynthesis was 0.9383 of
 ↪  today's at gsum = 0.06436 H_0; the slower expansion is equivalent to -0.379 neutrino species,
 ↪  and the helium fraction of standard nucleosynthesis falls from 0.24540 to 0.24008. If this is
 ↪  ignored (helium fraction 0.24540), the temperature spectrum changes by at most 8.6e-3 of
 ↪  itself for 30 <= l <= 2500, the Hubble constant at fixed angle by 0.045 km/s/Mpc, and chi^2
 ↪  against Planck-like mock data (other parameters fixed) from 468.3 to 335.3
[PASS] def-E8  (numerical) the growth of structure (file data/cmb_deflation_growth.csv). From a =
 ↪  0.01 to today, at k = 0.1/Mpc, the density contrast of matter in the Einstein-Boltzmann
 ↪  calculation grows by the factor that the growth equation of the same closure for pressureless
 ↪  matter gives: the ratios to the model without deflation agree to 1.3e-4 for the five rates and
 ↪  both closures, at the Hubble constant of each model and at a fixed Hubble constant; with the
 ↪  equation of the other closure they would disagree by at least 0.24 gsum/H_0. At a fixed Hubble
 ↪  constant the growth is 0.99856 (closure "Momentum") and 0.99933 ("Pressure") of its value
 ↪  without deflation at the rate of lunar laser ranging, 0.9712 and 0.9862 at the rate of
 ↪  nucleosynthesis; at a fixed angle, 1.00001 and 1.00077, and 1.0000 and 1.0160. The amplitude
 ↪  sigma_8 of the full calculation for the same primordial amplitude is 0.99892 and 0.99969, and
 ↪  0.9762 and 0.9918, of its value without deflation (0.82275): it contains, besides this growth,
 ↪  the change of the scale 8 Mpc/h with the Hubble constant and the evolution before a = 0.01
 ↪  with a weaker gravity
[PASS] ALL  (numerical) every check above passed
```

A.3 `deflation forecast`

```
PROJECT spacetimes, Revision_Deflation: checks of the forecast for the rate of deflation from mock
 ↪  data of the microwave background
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation forecast
5 of 5 checks passed

[PASS] def-F1  (numerical) the mock data and the Fisher matrices of the main project, computed
 ↪  with the copy of its program: the fiducial model has chi^2 = 0, and the Fisher errors of the
 ↪  six LambdaCDM parameters agree with the file data/dr_fisher.csv of the main project to 3.1e-11
 ↪  for both data sets
[PASS] def-F2  (numerical) forecast for the rate of deflation (Fisher matrix;
 ↪  data/deflation_fisher.csv). With the six LambdaCDM parameters free, Planck-like data
 ↪  determine gsum/H_0 to 0.01963 (one standard deviation; closure "Momentum") and 0.01974
 ↪  ("Pressure"); the upper bound at 95 per cent is gsum < 0.0385 H_0 = 2.65e-12 per year. Adding
 ↪  the deep survey gives 0.01056 and a bound of 0.0207 H_0. With the six parameters held fixed
 ↪  the error would be 0.00108: most of the effect of the deflation on the spectra can be absorbed
 ↪  by the other parameters. The bound lies between that of lunar laser ranging (0.00324 H_0) and
 ↪  that of nucleosynthesis (0.0644 H_0): 11.9 times weaker than the first and 1.7 times stronger
 ↪  than the second. Halving the step of the derivative changes the error by 8.2e-4
[PASS] def-F3  (numerical) the Hubble constant along the degeneracy: per unit gsum/H_0 the
 ↪  best-fit Hubble constant changes by -84.35 km/s/Mpc (Planck-like) and -82.11 km/s/Mpc (with
 ↪  the deep survey), -3.245 and -1.699 km/s/Mpc at the bound of 95 per cent: it FALLS. (At a
 ↪  fixed angle theta_* and fixed densities the command `cosmology` finds -65.24 km/s/Mpc per unit
 ↪  gsum/H_0 at small rates; in the fit the densities and the other parameters move as well.) The
 ↪  error of the Hubble constant grows from 0.561 to 1.748 km/s/Mpc when the rate is free.
 ↪  Deflation cannot raise the Hubble constant inferred from the microwave background
```

```
[PASS] def-F4  (numerical) fits at fixed rate (profile; data/deflation_profile.csv). For gsum up
 ↪  to 0.01 H_0 the minimum chi^2 is 1.00 to 1.01 times (gsum/sigma)^2 with the Fisher error
 ↪  sigma, and the shift of the best-fit Hubble constant 1.00 to 1.01 times the Fisher slope: the
 ↪  Fisher matrix describes the fits. The profile reaches chi^2 = 3.84 at gsum = 0.0376 H_0
 ↪  (Planck-like; Fisher bound 0.0385) and 0.0205 H_0 (with the deep survey; 0.0207). At the rate
 ↪  of lunar laser ranging the best fit has chi^2 = 0.027 and H_0 = 67.086 km/s/Mpc; at the rate
 ↪  of nucleosynthesis chi^2 = 11.6 (3.4 standard deviations) and H_0 = 61.70 km/s/Mpc
 ↪  (Planck-like), chi^2 = 39.8 (6.3 standard deviations) with the deep survey: the largest rate
 ↪  that nucleosynthesis allows would be excluded by such data. At every rate the best-fit Hubble
 ↪  constant is lower than without deflation
[PASS] def-F5  (numerical) accuracy of the fits. (i) Recomputing the derivatives of the spectra at
 ↪  the best fit for gsum = 0.02 H_0 and continuing the fit changes chi^2 by at most 0.000 and the
 ↪  Hubble constant by at most 0.001 km/s/Mpc (data/deflation_chord_check.csv). (ii) The closure:
 ↪  refitting with the closure "Pressure" at gsum = 0.01 H_0 and at the rate of nucleosynthesis
 ↪  changes chi^2 by at most 0.9 per cent of itself (or of 1, where it is smaller than 1) and the
 ↪  best-fit Hubble constant by at most 0.0005 km/s/Mpc: the forecast, which uses the multipoles l
 ↪  >= 30, does not depend on the closure
[PASS] ALL  (numerical) every check above passed
```

A.4 deflation families

```
PROJECT spacetimes, Revision_Deflation: checks of the forecast for the dark-radiation families of
 ↪  the main project on the deflating spacetime
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation families
4 of 4 checks passed

[PASS] def-N1  (numerical) the forecast of the main project for its dark-radiation families,
 ↪  repeated on the deflating spacetime at the rate that lunar laser ranging allows (gsum =
 ↪  0.00324 H_0). The data are the spectra of the fiducial model without deflation (chi^2 = 0 for
 ↪  that model). The deflating model without extra radiation fits them with chi^2 = 0.027 and H_0
 ↪  = 67.086 km/s/Mpc (Planck-like), chi^2 = 0.094 and H_0 = 67.094 km/s/Mpc with the deep survey
[PASS] def-N2  (numerical) the families on the deflating background against the main project (its
 ↪  files data/dr_fisher.csv and data/dr_profiles.csv): the Fisher errors of N agree to 0.2 per
 ↪  cent (Planck-like) and 0.3 per cent (with the deep survey) for the 7 families. In the fits at
 ↪  fixed N = 0.1 to 1 the best-fit Hubble constant is lower than in the main project by 0.274 to
 ↪  0.279 km/s/Mpc (Planck-like) and 0.266 to 0.270 km/s/Mpc (deep survey); the model without
 ↪  extra radiation is lower by 0.274 and 0.266 km/s/Mpc. The minimum chi^2 is LOWER than in the
 ↪  main project, by up to 3.11 and 8.06 at N = 1 (check def-N4 explains why). Deflation at this
 ↪  rate lowers every Hubble constant of the main project by about a quarter of a km/s/Mpc
[PASS] def-N3  (numerical) accuracy of the fits on the deflating background: recomputing the
 ↪  derivatives of the spectra at the N = 0.5 best fit of each family and continuing the fit
 ↪  changes chi^2 by at most 0.054 and the Hubble constant by at most 0.050 km/s/Mpc
 ↪  (data/dr_deflation_chord_check.csv). The scan of the transition redshift of the stepped fluid
 ↪  of the model at N = 0.4 (data/dr_deflation_zt_scan.csv) has its best fit at z_t = 10^3.00
 ↪  (chi^2 = 0.944); its Hubble constants are lower than those of the main project by 0.272 to
 ↪  0.277 km/s/Mpc
```

```
[PASS] def-N4  (numerical) the rate of deflation and the density N of extra radiation as free
 ↪  parameters together (Fisher matrices with eight parameters at the fiducial model;
 ↪  data/dr_deflation_joint.csv). (i) With the rate held at zero the errors of N are those of the
 ↪  main project (to 2.6e-11), and the marginal Fisher matrix of the pair reproduces the errors of
 ↪  each parameter alone (to 4.4e-16). (ii) N and the rate are positively correlated for every
 ↪  family: correlation coefficient 0.37 to 0.89 (Planck-like) and 0.28 to 0.93 (with the deep
 ↪  survey). Extra radiation makes the early universe expand faster, the weaker gravity of the
 ↪  deflating spacetime makes it expand more slowly, and the two compensate in part. With both
 ↪  free the error of gsum/H_0 grows from 0.0196 to between 0.0211 and 0.0429 (Planck-like), and
 ↪  the error of N by the same factor. (iii) At the largest rate that lunar laser ranging allows
 ↪  the best-fitting density is N = 0.005 to 0.023 instead of zero (chi^2 = 0.006 to 0.023 instead
 ↪  of 0.027). The difference between the minimum chi^2 of the fits on the deflating background
 ↪  and those of the main project at N = 0.1 and 0.2 is 0.99 to 1.01 times the cross term of the
 ↪  Fisher matrix. (iv) For every family and both data sets the largest Hubble constant with chi^2
 ↪  <= 3.84 (N >= 0, gsum >= 0) is reached at ZERO rate (a rise of 0.78 to 2.95 km/s/Mpc,
 ↪  Planck-like): the deflation lowers the Hubble constant faster than it makes room for extra
 ↪  radiation, and it cannot help the dark radiation to raise the Hubble constant
[PASS] ALL  (numerical) every check above passed
```

A.5 `deflation superfluid`

```
PROJECT spacetimes, Revision_Deflation: checks of superfluid dark matter on the deflating
 ↪  spacetime
generated by: deflation (Rust crate spacetimes-deflation, version 0.1.0; PROJECT spacetimes,
 ↪  Revision_Deflation); command: deflation superfluid
2 of 2 checks passed

[PASS] def-U1  (numerical) superfluid dark matter without deflation: the copy of the
 ↪  Einstein-Boltzmann program reproduces the files data/sfdm_summary.csv and
 ↪  data/sfdm_bounds.csv of the main project (half-mode wavenumbers of 15 models; limits on the
 ↪  fermion mass and on the sound speed of a condensate) to the digits written there (largest
 ↪  relative deviations 3.4e-11 and 3.1e-11)
[PASS] def-U2  (numerical) superfluid dark matter on the deflating spacetime
 ↪  (data/sfdm_deflation_summary.csv, sfdm_deflation_bounds.csv). The wavenumber below which the
 ↪  first sound of the fluid suppresses structure (half-mode) changes by -0.164 to -0.149 per cent
 ↪  at the rate that lunar laser ranging allows and by -3.28 to -3.04 per cent at the rate that
 ↪  nucleosynthesis allows (15 models; each background at the Hubble constant that keeps the angle
 ↪  theta_*). The limits that follow from the Lyman-alpha bound on thermal warm dark matter of 5.3
 ↪  keV move from m > 1.514 keV (fermions, weak pairing) and c_s0 < 1.645e-11 c (condensate;
 ↪  Thomas-Fermi radius < 0.365 pc) without deflation to m > 1.516 keV, c_s0 < 1.641e-11 c (0.364
 ↪  pc) at the rate of lunar laser ranging and m > 1.555 keV, c_s0 < 1.565e-11 c (0.347 pc) at the
 ↪  rate of nucleosynthesis. The suppression is set while the fluid still has pressure, long
 ↪  before today, when gravity was weaker: the Jeans wavenumber is proportional to the square root
 ↪  of Newton's constant, which in the early universe was 0.9969 and 0.9383 of today's value at
 ↪  the two rates (square root minus one: -0.154 and -3.13 per cent). The two closures give the
 ↪  same half-mode wavenumbers to 1.5e-5
[PASS] ALL  (numerical) every check above passed
```

Appendix B. Files

file	content	written by
`code/scripts/vendor_eb.py`	writes the four copied modules from the sources of the main project	—
`code/deflation-rs/src/eb_cosmo.rs`, `eb_boltzmann.rs`, `eb_spectra.rs`, `eb_fit.rs`	the Einstein–Boltzmann program on the deflating spacetime (copies; not edited by hand)	`vendor_eb.py`

file	content	written by
code/deflation-rs/src/spectra_defl.rs	assumption A3 in words; shared routines; nine unit tests	—
code/deflation-rs/src/cmd_spectra.rs	the commands background, spectrum, forecast, families, superfluid	—
code/deflation-rs/src/cosmo_defl.rs	the growth equations of the closures	—
data/background_deflation.csv	Sec. 5.2	deflation background
data/hubble_deflation.csv	Sec. 6	deflation cosmology
data/growth_deflation.csv	Sec. 4	deflation matter
data/cmb_deflation_summary.csv, cmb_deflation_cl.csv, cmb_deflation_pk.csv, cmb_deflation_peaks.csv	Secs. 7.1, 7.2	deflation spectrum
data/cmb_deflation_closure.csv, cmb_deflation_gaps.csv	Sec. 7.3	deflation spectrum
data/cmb_deflation_convergence.csv, cmb_deflation_growth.csv	Secs. 7.4, 7.5	deflation spectrum
data/deflation_fisher.csv, deflation_profile.csv, deflation_chord_check.csv, deflation_residuals.csv	Sec. 8	deflation forecast
data/dr_deflation_fisher.csv, dr_deflation_profiles.csv, dr_deflation_chord_check.csv, dr_deflation_residuals.csv, dr_deflation_zt_scan.csv, dr_deflation_joint.csv	Sec. 9	deflation families
data/sfdm_deflation_transfer.csv, sfdm_deflation_summary.csv, sfdm_deflation_bounds.csv	Sec. 10	deflation superfluid
data/background_checks.txt, spectrum_checks.txt, forecast_checks.txt, families_checks.txt, superfluid_checks.txt (and .csv)	Appendix A	the five commands
data/deflating_checks_independent.txt	checks I44d-P1, I71d-P1, Idef-P2 to Idef-P8, Idef-C2	verify_deflating_independent.py

file	content	written by
`data/mutation_test_independent.txt`	sixteen deliberate errors in the independent verification, and the check that detects each (provenance RP8, Sec. 4)	`mutation_test_independent.py`
`figures/fig_spectra`, `fig_closure`, `fig_forecast`, `fig_families`, `fig_sfdm` (`.pdf`, `.png`)	the five figures	`make_figures.py`

References

1. C.-P. Ma and E. Bertschinger, *Cosmological perturbation theory in the synchronous and conformal Newtonian gauges*, Astrophys. J. **455**, 7 (1995).
2. P. J. E. Peebles, *Recombination of the primeval plasma*, Astrophys. J. **153**, 1 (1968); S. Seager, D. D. Sasselov and D. Scott, *A new calculation of the recombination epoch*, Astrophys. J. **523**, L1 (1999).
3. Planck Collaboration, *Planck 2015 results. XIII. Cosmological parameters*, Astron. Astrophys. **594**, A13 (2016), Eq. (67) (fit to the code PArthENoPE).
4. M. Viel, J. Lesgourgues, M. G. Haehnelt, S. Matarrese and A. Riotto, *Constraining warm dark matter candidates including sterile neutrinos and light gravitinos with WMAP and the Lyman-α forest*, Phys. Rev. D **71**, 063534 (2005).
5. The supplement S3 of the main project and its references; the supplement RS3 of the revision and its references (bounds on the variation of Newton's constant).